

The Standard Model as a Constraint-Derived Carrier–Response Structure

Joint fermionic–bosonic classification, shared-source coherence, and physical projection

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Abstract

We construct a structural Standard Model atlas in the AASC constraint framework, distinguishing premetric source identity, standing response and physical realization. The independently established fermionic three-root structure is retained, while its current source-role incidences and the bosonic response laws are assembled over one canonical gauge–representation–charge–Higgs source. Whole-source reindexing preserves relative flavor and CP information without choosing a preferred generation labeling or identifying incompatible physical preparations. A domain-sensitive mixed-use algebra yields the greatest observational congruence for the declared target, while complete source reconstruction gives factorization of every independently admitted covariant use. On a represented local-law branch, tensor probes recover the charged interaction coefficients. We further prove that complex traces of words of length at most seventeen classify a pair of three-dimensional Hermitian quark Gram matrices up to simultaneous unitary conjugacy, including singular and reducible cases. An independently specified structural law-role demand has positive simultaneous occupancy and exact fixed-datum closure. Conventional particle labels factor through typed projection and marking records; no additional same-scope carrier identity is supplied by nomenclature. The atlas also retains the premetric gravitational constraint role, exact process domains and the distinct authority of stronger physical projections. The result is a branch-complete structural classification and reconstruction theorem, not a selection of numerical parameters or an unrestricted common physical realization.

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1 The shared structural target

A particle table organizes familiar physical designations, but it does not by itself distinguish the structural source of a particle role from its response to other sectors or from the conditions of its physical realization. A constraint-derived presentation must make those distinctions explicit. In particular, it must preserve the incidences through which matter responds to gauge transport, the shared Higgs source, and the relative information observed by a joint process. Adding separate inventories or taking a Cartesian product of their observable values does not establish these common-source relations.

The fermionic analysis establishes the positioned occurrence and Higgs-response architecture and its exact terminal distinctions [2]. The bosonic analysis establishes source-law and response reconstruction, global gauge–Higgs coherence, and physical projections on their actual domains [6]. The present paper combines these results into a carrier–sector–response–projection classification. Its central distinction is between a complete source law and a particular argument of that law. Uniqueness of the former does not imply uniqueness of a response value, and a response need not have an independently realized particle pole.

Here a *source* comprises the actual structural laws and their witnessed incidences. A *target* fixes the questions, admissible operations, comparison class and retained boundaries. A *mark* records an actual operand or preparation, not a preferred representative selected from an equivalence class. A physical projection has the *authority* of the mathematical construction supporting it, with that construction’s hypotheses retained. The word does not refer to institutional endorsement.

1.1 The inherited fermionic and bosonic structures

Write $F = \{N, L, U, D\}$ for the four received matter kinds. The established root object is $I_3 = \text{Occ}_N$, with three elements and an external S_3 action. For a positioned kind r_f , the canonical source of its incidence map is

$$E_f = I_3 \times \{r_f\}, \quad \text{Disc}(E_f) \xrightarrow{\eta_f} \mathcal{S}_f(C_f) \xrightarrow{\chi_f} Q_f^Y.$$

These are the source-defined arrows themselves, not newly chosen bijections. Their primitive-standing and complete-occupation receipt precedes any use of familiar generation names or matrix dimensions [21, Theorems 6.2 and 5.4][1, Sections 3–4]. The primitive circuit rank is one only after this architecture has been established [21, Corollary 5.5]. The projection $p_f : E_f \rightarrow I_3$ induces the convenient fixed-kind composite $\eta_f \text{Disc}(p_f^{-1})$; it does not change the source of η_f [2, Sec. 1 and Thm. 10.14].

The bosonic import retains the complete charge, weak, and color law families and their actual source charts, together with one canonical Higgs bridge. The bridge is the full tuple

$$\mathcal{B}_H = (W_{\text{br}}/\sim_{\text{br}}, \Sigma_{\text{br}}, \beta_{\text{red}}, \rho_{\text{br}}), \quad C_H = H_{\text{br}/\text{fix}},$$

including its actual order, guard and relative-incidence data. Equality of its name is not a replacement for equality of this tuple. The fermionic incidence and the bosonic source-input operation use it through distinct maps

$$\beta_{F,f} : Q_f^Y \longrightarrow \mathcal{H}_{F,f}(C_H), \quad \beta_{H,EW}^B : C_H \times D_{EW} \rightsquigarrow \text{Fix}_{EW}.$$

The first attaches a complete rooted incidence record to an established response slot. A many-input presentation of its source graph is not an inverse of that arrow. The second has the actual dependent source domain specified by its bosonic construction; it is distinct again from the later metric projection. All sibling inputs and attachments are retained [2, Defs. 5.1–5.2][6, Thm. 5.4].

The bosonic whole-law inventory $(1, 1, 1; 1)$, the fermionic positioned roots, the later role-demand set, and physical particle projections count different objects. We do not add their

cardinalities to obtain a carrier census. The source semantics and the common-target role problem must be formed first.

1.2 Actual joint diagrams

Definition 1.1 (Shared structural target). A target T fixes the received source theories; the full Higgs bridge identity; matter kind, charge, chirality and color incidences; the allowed primitive mixed operations with their independently established partial graphs; source and boundary comparisons; upstream continuation; primitive probes and failure diagnostics; and the authority layer of every requested projection. Its source domain X_T consists of actual finite witnessed source diagrams formed from these theories and operations. It retains the source charts and all interface maps joining them.

A joint diagram has distinguished restrictions r_F and r_B to the received fermionic and bosonic subdiagrams and their witnessed shared-source identifications. No general diagram is required to be a restriction of one universally amalgamated pointed record. A lawfully shared diagram is a single joint object even when its internal charts remain distinct. A tuple of unrelated marginal quotient classes is not a source diagram.

The admitted primitive graphs and their parameters are fixed before the observation quotient. A new independent operator, changed boundary, new source mechanism or stronger physical authority defines a declared target extension. Its reception and the effect on the earlier quotient are separate proof obligations.

Three mathematical obligations remain distinct in this definition: evaluation of the source diagrams, coverage of their admitted uses, and existence of compatible diagrams at a proposed target. A formal fibre product of response signatures proves none of these by itself. The following sections establish evaluation and coverage from the actual source operations, then construct a common structural source with the full shared compatibility data. The resulting quotient therefore follows the source construction rather than licensing it.

When a target has a global group, bundle, metric, parameter point, representation, state or asymptotic attachment, write $b(x)$ for those retained coordinates and

$$B_T = \text{im}(b : X_T \rightarrow \mathcal{B}_T), \quad X_{T,b} = b^{-1}(b).$$

The base is an actual source image. A formal coefficient range need not equal it. Restricting several conclusions to $X_{T,b}$ uses the same values and attachments, not separately fitted sector parameters.

1.3 Preservation of the inherited structures

Proposition 1.2 (Preservation of the accepted structural chain). *Receiving a joint source diagram through its distinguished predecessor restrictions preserves the canonical positioned fibres, η_f , χ_f , the full Higgs source, and every retained predecessor boundary. Erasing the newly adjoined vertices returns those same objects and arrows. A new mixed theorem cannot be presented as a stronger statement of an imported theorem.*

Proof. The distinguished restrictions retain the literal source objects and maps, not only their isomorphism types or numerical dimensions. Deleting the new vertices changes none of their incidences. A change of source presentation transports the entire retained diagram and therefore gives the same predecessor theorem through its lawful naturality map. The fermionic handoff requires a new premise or inference rule for a genuinely stronger old-language conclusion [2, Thm. 10.14]. Such a rule is part of the new mixed source construction, not part of the erased

diagram. The same restriction discipline holds for the bosonic handoff [6, Prop. 3.7 and Thm. 11.5]. $\square$

The charged and colored terminal dispositions in FCGS-III are not positive terminal witnesses, and its neutral positive objects retain their certified child scopes. The absence of an admitted terminal witness on that source domain is not a proof of physical nonexistence. A stronger realization theorem can extend the available domain only by supplying its additional source witnesses; it does not change the original result.

1.4 Structural governance and realization

Nondegenerate determinate source objects are governed by $\mathcal{K} = (\text{Adm}, \text{Stand}, \text{Ref}, \text{Irr})$. The kernel is not an optional extra premise applied only to favorable realizations. In the regular AER comparison class the premetric gravity theorem identifies the constraint, redescription and relational localization role [18]. This role is distinct from the carrier-law families and is not an additional Standard Model carrier. A metric and its associated field equations enter only through their licensed source and realization maps.

We keep three questions distinct. Structural classification concerns the laws, source incidences, rooted sectors and complete response comparisons. Classical realization adds one source-attached metric history obeying the coupled equations on its specified branch. Physical channel realization adds the particular state, operator, spectral, pole, confinement or asymptotic constructions needed by the claim. The second and third questions can refine a structural description without creating its earlier sources.

The BCRS-III classical construction provides a populated gauge-Higgs-gravity branch with a retained scalar clock and the full graded matter carrier. Its chosen fermion fields vanish. It is thus not used here to establish the fermionic occurrence witnesses or to infer simultaneous nonzero population of all matter responses [6, Sec. 7]. The structural common-source proof instead retains those witnesses explicitly.

2 Source reception and operational completeness

Two distinct completeness questions govern the construction. The carrier classification must cover the source and status claims at its declared role granularity; the observation algebra must cover every admitted use of those sources. The first does not imply the second: a closed carrier inventory need not generate all mixed observables. The operations are therefore specified independently of the quotient, and their finite-network normal form is then proved. Role demand and primitive surplus require separate arguments, developed below.

2.1 Primitive graphs before signatures

Definition 2.1 (Received operational target). For every admitted source mechanism let g_λ be its actual partial graph

$$g_\lambda : D_\lambda \subseteq X_{s_1} \times \cdots \times X_{s_k} \longrightarrow X_{s_o}.$$

Its source record includes the complete operand sorts, intrinsic structural parameter λ , common-interface maps, domain predicate, output type and source-defined failure diagnostic. A parameter can be an entire operator or functional when that is the independently supplied source object. It cannot be a table of desired response values substituted for an absent operation.

The structural operational families are:

- (i) the actual fermionic occurrence, root, component, response-slot and Higgs-incidence maps, their restrictions and lawful transports;
- (ii) the gauge transition, representation, charge, loop, obstruction, product, residual and color-support operations of the bosonic source;
- (iii) full bridge-witness, order, deformation and fixation operations, with their actual dependent mixed inputs;
- (iv) mixed representation action, charge/current evaluation, invariant tensor contraction, adjoint or conjugation, and source-admitted coupling and relative-flavor operations;
- (v) boundary extraction, source restriction, independently formed continuation and prefix tests, activity and failed-law diagnostics.

At a stronger target the family additionally receives the actual metric/source, state, operator, spectral, asymptotic, composite or record graphs whose source hypotheses are supplied. Their potentially infinite-dimensional arguments and limit prescriptions are retained. Receiving such a graph does not assert its domain is populated.

A use of this source-generated operational target is an application of one of these native source graphs, an actual source restriction or comparison, or a finite well-typed composition of such applications using witnessed interfaces. An independently defined infinitary native operation is received as that operation, not inferred to be a finite composition or a finite-jet function. New independently admitted operations extend this target explicitly.

This target is fixed by the source laws and their uses, before the response quotient and without choosing a desired carrier count. The primitive reception has concrete content. The first family consists of the FCGS-III incidence maps and source domains. The second and third use the complete source-law network families of BCRS-I. The fourth requires joint-incidence and tensor checks: a bosonic theorem alone does not establish admission of an arbitrary fermionic sibling. Source-represented physical functionals in the last family keep their original operator and domain, even if their source definition involves completion or an infinite sum [3, Secs. 12–13][6, Secs. 3, 8–10].

2.2 Covariance with fixed intrinsic siblings

Lemma 2.2 (Relative transport at a new input port). *Let a primitive joint graph have an actual pointed source record. Suppose $u : x'_j \rightarrow x_j$ is a licensed invertible change of presentation of its j -th port. Re-present every outgoing incidence m as mu , every incoming incidence l as $u^{-1}l$, and every port endomorphism a as $u^{-1}au$. If the primitive source law is covariant on this complete diagram, replacing this one port preserves and reflects admission, output and failure while the other intrinsic inputs remain fixed. The same assertion holds for a certified compression only when all incident maps and domains have relative descent to its common source record.*

Proof. The displayed substitutions make each incident square commute. They change a representation of the attachment, not its intrinsic value. The unchanged sibling ports therefore meet the same intrinsic input record. Source covariance preserves its domain predicate and its output and failure records. Applying the inverse comparison proves the reverse domain implication. For a compression, both presentations descend with all incident maps to the certified common record; equality there replaces inversion of a raw quotient map. Without that relative descent the conclusion does not follow. This is the complete pointed comparison used by intrinsic source descent, now applied to each explicitly received new port [3, Sec. 12]. $\square$

Proposition 2.3 (Primitive fermionic and mixed reception). *The received fermionic incidence maps and source-admitted mixed tensor operations satisfy the relative transport lemma. Their intrinsic partial graphs are independent of the chosen source presentations. No new full-target arrow between distinct fermion kinds is thereby created.*

Proof. For a lawful same-kind comparison $u : X \rightarrow X'$, write e_u, s_u, a_u, h_u for its maps on roots, components, response slots and complete Higgs-incidence records. The accepted naturality equations are

$$s_u \eta_X = \eta_{X'} e_u, \quad a_u \chi_X = \chi_{X'} s_u, \quad h_u \beta_{F,X} = \beta_{F,X'} a_u.$$

They hold on the complete target-preserving groupoid, whose inverse comparisons retain the same source, continuation, boundary and failure fields [2, Thms. 3.2 and 5.4]. The actual bridge projection of the incidence record commutes with the full bridge comparison. Transporting an additional incident map by Lemma 2.2 therefore keeps its fixed sibling intrinsically unchanged. Only arrows already licensed at that kind are used.

For a mixed multilinear tensor $M : V_1 \otimes \cdots \otimes V_k \rightarrow W$, a presentation change U_j at one input and U_o at the output gives

$$M' = U_o M(1, \dots, U_j^{-1}, \dots, 1), \quad M'(v_1, \dots, U_j v_j, \dots, v_k) = U_o M(v_1, \dots, v_j, \dots, v_k).$$

For an antilinear slot use its conjugate representation; the same calculation then applies to the corresponding linear conjugate-space slot. Representation action satisfies $\rho'(a) = U\rho(a)U^{-1}$, and adjoints and contractions transport with the retained Hermitian or duality structures. These identities prove covariance of the actual representation/current/coupling graphs, not invariance under a physical variation of one operand. Their additional standing-admission predicates are the retained source witness laws and are transported with the entire record. Algebraic invariance alone is not substituted for a missing coupling witness. Inverse transport reflects those predicates. The mixed graphs thus satisfy the lemma before any observational quotient is formed. $\square$

For example $z = \ell^* Y r$ is unchanged under $r' = U r$, $Y' = Y U^{-1}$, with ℓ fixed. Varying r while keeping the physical Y fixed generally changes z . The latter is a new response, not a source-skin comparison.

2.3 Network normal form

Definition 2.4 (Source-decorated mixed network). A mixed network is a finite typed diagram of the received primitive graphs. Edges carry actual operands and their interface maps. Vertices carry their source parameters and admission witnesses, never an independently chosen output. Subdiagrams retain their original charts. The evaluation order is part of the source use when partial operations make order significant. A source-given order-independent convention can replace it only with its corresponding equality proof.

Theorem 2.5 (Operational reception and completeness). *Every use of the independently fixed operational target of Definition 2.1 has a source-decorated mixed network with exactly the same intrinsic input, output, definedness, continuation, boundary and failure information. Conversely every such evaluated network is a use of that target. Full-diagram lawful presentation changes preserve its evaluation and reflect its domain.*

This is completeness for the received source-generated operational target. It does not certify an unlisted independent operator, identify an infinitary functional with a finite polynomial, or assert that all formal marginal tuples are source-realizable.

Proof. At a native operation application, take its actual source graph and the operand and parameter records supplied by that application. The old bosonic cases are precisely their source-law constructors [3, Def. 13.5 and Thm. 13.9]. The fermionic and mixed cases are received

by Proposition 2.3 and their individual source-admission checks. A restriction uses the actual restricted operand, a comparison transports its entire pointed chart, and a continuation application retains its source segment and exact domain. A represented state or spectral functional remains the same primitive graph, with its complete arguments. Its graph is not replaced by finitely many sampled values.

For composition, graft the already constructed networks along the given common-interface maps. The original source expression guarantees that these are the actual maps in that expression; equality of marginal root names is not used as a substitute. Primitive evaluation therefore computes the same next value on the same domain. Induction over the finite construction yields the complete network. A shared input is retained as one edge with several incidences, so the induction does not independently rechoose it at different vertices. Conversely, reading a network in its declared evaluation order gives the native applications, restrictions and compositions from which it was formed.

Every old primitive is source-covariant and every new primitive satisfies the relative transport lemma. Induction consequently preserves the domain of each vertex as well as its output. At a failed vertex the retained source diagnostic is unchanged. Thus the complete partial evaluation descends and reflects its domain. The proof is an induction on finite source use, not on the cardinality of a response signature or on a desired Standard Model inventory. $\square$

2.4 Exact failure and the boundary of coverage

For a partial probe $p : D_p \rightarrow V_p$, use the totalized observation

$$\hat{p}(x) = \begin{cases} (\text{defined}, p(x)), & x \in D_p, \\ (\text{undefined}, \delta_p(x)), & x \notin D_p. \end{cases}$$

Here δ_p is the source-defined diagnostic; if no finer diagnostic is supplied, the undefined branch has a single formal value. Each primitive operation's failure diagnostic is itself a received observation on its complete input record. Sequential composition records the first failed vertex and that diagnostic. It does not inspect an unexecuted sibling and append new information after the quotient. A defined negative, an inactive response, a not-applicable mechanism and **NotReady** are separate values, not alternative names for undefined execution.

Proposition 2.6 (No silent operational extension). *Suppose x, y agree on the complete received observation family, but an independently proposed operation has different domain or observable value on them. That operation cannot descend through the old quotient. If it was already licensed by the fixed source and target, its omission refutes the claimed coverage and that proof must be reopened. If it instead requires a genuinely additional operator, domain or authority, its source-proved extension refines the observational relation. Neither case thereby changes the earlier source-formed occurrence theorem.*

Proof. A function on an equivalence class has one definedness value and one totalized output. The proposed difference contradicts their being functions of the old class. An already licensed omitted test contradicts completeness of the claimed old family. For a genuinely additional test, adding its source-proved graph and all its contexts adds tests, so the extended equivalence is contained in the old one. The accepted occurrence objects and their incidence maps remain literal restrictions by Proposition 1.2. No reverse constructor from the new output into those objects has been introduced. $\square$

The completeness claim thus has a precise falsifier: an omitted use already licensed by the fixed target. Such an omission refutes coverage; it is not a change of target. Only an additional source operation, domain or authority extends the target. The distinction is determined by the source and its declared use domain, not by the particle name assigned to an output.

3 Continuation before physical projection

A continuation test need not presuppose a particle trajectory, a solved field history or an evolution operator on a Hilbert space. The structural quotient uses an independently admitted source continuation, retaining its orientation, boundary and failure information. Physical processes arise through later projections or additional typed operations; they do not define the source continuation itself.

3.1 The source continuation domain

The source target retains its native continuation graphs before the mixed signature is formed. Their domain is supplied by the original source, not by the predicate that a desired final projection succeeds. Restrictions and compositions carry their actual source maps. When several source charts are joined, a simultaneous continuation consists of source continuations and their compatible interface maps. It is not obtained by selecting separately solved physical histories that happen to have matching endpoints.

An explicit local construction uses the upstream part of GRQM Module C [17, Secs. 4–11]. It begins with a regular local chart U of the realized physical interior, its standing and failure data, a certified local clock representative τ , and an independently supplied locally Lipschitz continuation seed V . The seed is tensor content in an already admitted interior. The kernel neither selects it nor derives its numerical values. The source normalization is $d\tau(V) = 1$. The chart and seed are formed without metric, Hilbert, record or compatibility *projection maps*; this does not delete their upstream standing and record-order data.

Definition 3.1 (Continuation-ready source interface). A continuation-ready interface is an actual source attachment of the structural target to a Module C chart and seed, retaining the source maps for its roles, boundaries and incidence records. Its permitted segments remain inside the declared local source domain and satisfy the complete standing and no-repair conditions there. The attachment is supplied before its physical projection maps. An arbitrary structural diagram is not declared to possess such an attachment.

Theorem 3.2 (Projection-independent local continuation). *On every continuation-ready interface, the independently supplied seed constructs an admitted clock-forward Moore-history category. It has identities, prefix restriction and endpoint-compatible composition. Deleting the later metric, quantum, record and compatibility projections leaves this category and its source attachments unchanged. Where an admitted nonzero seed has a local standing interval, the category contains a nonidentity segment. No failed prefix becomes an admitted history by appending a later correction on the same lineage.*

Proof. In a source chart local existence and uniqueness for the locally Lipschitz V give the local integral curves $\alpha_t(x)$. The Module C source theorem supplies this construction before projection [17, Thm. 7.1]. Uniqueness gives $\alpha_{s+t}(x) = \alpha_s(\alpha_t(x))$ on its exact common domain, and integration of $d\tau(V) = 1$ gives

$$\tau(\alpha_t(x)) = \tau(x) + t.$$

Restrict to segments for which every point remains in the admitted standing domain, with all boundary and failure conditions satisfied. The restriction is substantive: no claim of global existence or passage through a standing boundary follows from local tangency. An admitted segment is (x, t) , with $t \geq 0$, source x and target $\alpha_t(x)$. Composable segments satisfy

$$(\alpha_t(x), s) \circ (x, t) = (x, t + s).$$

Addition of durations and uniqueness give strict associativity; $(x, 0)$ is the identity. The actual source restriction maps transport the retained structural records along these segments. Lawful full-record presentation comparisons preserve the same durations, domains and endpoint incidence squares, hence descend exactly as in the source's fixed-clock category [17, Thm. 11.1].

Every object used in this construction is upstream chart, clock, seed, source attachment, standing, boundary or failure data. None is a metric solution, a quantum propagator or the value of a later record/compatibility projection. Deleting those maps therefore leaves the construction intact [17, Cor. 11.2]. On a positive sufficiently small standing interval the clock increases, so its endpoint cannot equal its initial point. Finally an admitted history must have every prefix admitted. The source irreversibility condition forbids a later extension from changing the original failed-prefix status [17, Thm. 8.4]. $\square$

The theorem constructs continuation on the interfaces carrying the stated source data. Its local analytic chart does not exhaust the structural source image, and a chart need not support every physical realization. Outside this local envelope, native source continuations retain their independently established graphs. No prior metric realization is imposed on those graphs.

3.2 Clock changes and retained identity

Proposition 3.3 (External duration transport). *If $\tau' = f \circ \tau$, where f is a source-certified orientation-preserving local C^1 diffeomorphism, set $V' = (f' \circ \tau)^{-1}V$. The separately constructed categories are equivalent through*

$$(x, t) \longmapsto (x, \rho_f(x, t)), \quad \rho_f(x, t) = f(\tau(x) + t) - f(\tau(x)).$$

The equivalence preserves oriented source histories and their standing/failure and attachment records. It is not an identification of unequal durations inside a single fixed-clock category.

Proof. Define the new curve directly by

$$\alpha'_{t'}(x) = \alpha_{f^{-1}(f(\tau(x)) + t') - \tau(x)}(x).$$

The chain rule makes this a V' -curve on the transported interval. Conversely, the inverse time change converts any such curve to a V -curve, so uniqueness follows from the original source flow. No new local Lipschitz claim about V' is required for a merely C^1 clock change. For composable segments,

$$\rho_f(x, t + s) = \rho_f(x, t) + \rho_f(\alpha_t(x), s),$$

so source, target and composition are preserved. The inverse is the same construction for f^{-1} . Source-certified clock change preserves all non-duration records. This is the actual duration transport of Module C, not a quotient by a chosen time variable [17, Thms. 7.4 and 10.5]. $\square$

Similarly, an external permutation of the three occurrence roots does not identify them as one object. It re-presents the entire positioned incidence structure equivariantly. A source continuation can change a response or relate endpoints while the coexisting rooted occurrences remain distinct in their received standing object.

3.3 Descent of a continuation observation

Proposition 3.4 (Exact continuation-fibre criterion). *Let P be a source-defined profile on a continuation-ready interface. The local continuation descends through P , including its exact interval domain and failure information, if and only if those domain/failure records and the value $P(\alpha_t(x))$ are constant on every actual P -fibre for each requested t . On that image the descended process is unique and satisfies identity and composition wherever defined.*

Proof. A descended partial map must assign a unique definedness record, failure diagnostic and output to each $P(x)$, proving necessity. For sufficiency define its domain as the image of the original domain. Fibre constancy makes that domain saturated and the output independent of representative. The source identity and composition equations descend because their evaluations agree on every actual representative. Surjectivity onto $\text{im } P$ proves uniqueness there. No extension to unrealized profile values is defined. $\square$

The criterion tests descent of an independently defined continuation, rather than defining continuation through the success of a *later* profile. In particular, a profile that forgets a standing boundary can fail domain reflection even when two interior evolutions agree over a shorter common interval. The complete signature constructed next includes all requested continuation and failure uses; its descent is consequently established for those actual uses.

4 The joint observational quotient

Source equivalence identifies lawful presentations of the same witnessed object. Observational equivalence identifies source objects that no admitted use of the target distinguishes, and can therefore be coarser than source identity. Adding mixed uses, however, can refine the separate fermionic and bosonic observation quotients. These relations are determined by source comparisons and observations, not by particle names or representation dimensions.

4.1 Contexts, reduction and congruence

For a source sort s , let $\mathcal{C}_T(s)$ contain all one-hole contexts obtained from the received operational target. The hole can appear at any input port; all intrinsic sibling inputs, relative attachments and source parameters are fixed. Include every source-admitted continuation application. The contexts are formed before any equivalence class is counted.

Let p range over the primitive probes with matching output sort, including domain, activity and failure probes. Put

$$\Sigma_s(x) = (\widehat{p \circ C}(x))_{C,p}, \quad x \simeq_T y \iff \Sigma_s(x) = \Sigma_s(y).$$

Values are compared in their intrinsic output sorts, with actual transport of coordinate presentations. The operation-completeness theorem, not this signature definition, establishes coverage of the received target. Write Σ_T for the signature at the joint-source sort and $Q_T = \text{im } \Sigma_T$.

Theorem 4.1 (Greatest complete mixed congruence). *The relations $\simeq_T$ form the greatest many-sorted congruence preserving all primitive totalized observations and reflecting all primitive partial-operation domains. Replacing one input while fixing its intrinsic siblings preserves and reflects admission, output class and retained failure. Every received operation descends uniquely on its exact quotient domain. Source-presentation comparisons lie in this congruence.*

Proof. Signature equality is an equivalence relation. To replace one input of g , make g with its fixed siblings a one-hole context. Its domain and failure tests are primitive observations of that context, so they agree on equivalent inputs. When it is defined, compose any output context and probe after g . This is another received context of the original input; its equal evaluations show that the two outputs are equivalent. Replacing finitely many inputs one at a time proves the corresponding many-input assertion. Exact common-interface admission is tested at each step; it is not inferred from equality of unrelated marginal signatures.

Now let R be any many-sorted congruence preserving those primitive values and domains. Induction through a source context preserves definedness and R -related outputs. At its first

failed vertex the primitive failure observation agrees; otherwise the final probe agrees. Thus every context gives the same totalized value and $R \subseteq \simeq_T$. Quotient descent is therefore well-defined, with saturated domain and representative-independent output, and surjectivity onto the actual quotient image gives uniqueness. Finally source comparison preserves all primitives by source reception, including the new fixed-sibling mixed graphs. Context induction places those comparisons in $\simeq_T$. $\square$

Proposition 4.2 (Activity reduction without loss of output types). *Source-null or disconnected latent inputs are removed by the fixed source activity predicate, not by inspecting the cardinality of Q_T . Activity is constant on each complete-signature class. The active classes form distinguished subsets of the full quotient sorts, but need not form a subalgebra. Operations on active inputs retain their full output sort unless restricted to the exact active-output subdomain.*

Proof. The activity predicate is a received observation, so equivalent inputs have identical activity. Quotienting its true locus is therefore independent of representative. Nothing in the primitive source laws says that an operation with active inputs must have an active output; destructive interference or a vanishing contraction can give zero. Its full output class remains defined. If an active output is required, use the inverse image of the active-output subset. Congruence makes that subdomain saturated. No count is used in any of these constructions. $\square$

4.2 What the separate atlases omit

Pull back the complete predecessor signatures and all independently observed common-source fields to X_T . Denote their tuple by

$$m(x) = (\Sigma_F(r_F x), \Sigma_B(r_B x), \Sigma_J(x)).$$

Let μ contain the remaining mixed contexts fixed by the received source operations, including domains and failures. The indexing family is not chosen after looking at particular pairs of representatives.

Theorem 4.3 (Exact joint refinement and reconstruction). *On the actual joint source domain,*

$$\ker \Sigma_T = \ker m \cap \ker \mu.$$

The separate-atlas tuple reconstructs the full observational quotient if and only if μ is constant on every actual m -fibre. Equivalently, there is a unique map on $\text{im } m$ with $\mu = \bar{\mu}m$. A difference in a mixed domain, failure or value on one such fibre obstructs the reconstruction.

Proof. Operational completeness partitions the full test family into the received marginal/shared tests and the remaining mixed tests. Agreement on all tests is therefore exactly agreement on both tuples. Fibre constancy is necessary for a function of m . If it holds, define $\bar{\mu}(m(x)) = \mu(x)$; this is independent of the actual representative and recovers the full signature. Every point of $\text{im } m$ has such a representative, giving uniqueness. No value is assigned to an unrealized tuple outside that image. $\square$

In particular, the bosonic congruence need not be the full fermion–boson congruence: fixed-sibling mixed tests may distinguish marks that the separate atlas identifies. The predecessor theorems remain valid on their restrictions, while the joint quotient retains the additional distinctions. Section 7 establishes concrete finite reconstruction and separation results on the admitted mixed coefficient branch.

4.3 Source comparison and actual relative attachments

Let $\mathcal{X}_T$ be the groupoid of actual joint diagrams and their licensed complete comparisons. Let R retain their predecessor restrictions and all additional relative attachments, mixed tensors, domains and guard fields of a declared chart. The codomain below is its actual image, with arrows transported from those complete field records, not an ambient product of unrelated comparison groups.

Proposition 4.4 (Exact source-reconstruction criterion). *If the retained fields reconstruct each diagram and its defining incidence maps, and every comparison between those field records lifts to a licensed full-diagram comparison, then R classifies the actual source objects up to source equivalence. Without comparison reflection it need not do so, even if every object is covered. The observational quotient equals this source classification precisely when the complete observations separate its distinct actual orbits.*

Proof. A full comparison preserves every retained field. Conversely, by reconstruction a field-record isomorphism identifies all vertices, arrows, domains and guards of the diagrams, and the lifting hypothesis makes that identification lawful. Thus two objects have equivalent records exactly when they are source-equivalent. Object coverage alone supplies no such lifted arrow. Source comparisons already lie in the observational relation by Theorem 4.1; equality of the two classifications is therefore exactly the reverse separation condition on actual source orbits. $\square$

A convenient double-orbit expression for relative attachments requires actual attachment coverage, exact domain invariance, lifting of the acting comparisons and completeness for all actual comparisons. Otherwise the appropriate object is the actual relative-attachment groupoid itself [6, Thms. 4.2–4.3]. A larger formal automorphism group cannot be used to erase a mixed phase, a boundary charge or a source incidence that its elements do not preserve.

Corollary 4.5 (Conservative change of observational target). *For an actual source restriction $r : X_{T'} \rightarrow X_T$ that retains every T -observation with its domain, the new quotient has a unique induced map to the old quotient on $r(X_{T'})$. It is injective exactly when all new observations factor through the pulled-back old signature, and it reaches the whole old quotient exactly when the restriction meets every old observational class.*

Proof. Equality of the new signature implies equality of every pulled-back old test. This gives the quotient map. Injectivity is equality of the two kernels on $X_{T'}$, equivalently the stated fibre factorization. Surjectivity is exactly the stated actual-image condition. Neither assertion follows from having similar notation. $\square$

5 A common Higgs source and simultaneous structural compatibility

At structural resolution, the matter incidences and bosonic source network admit a constructive common diagram through their canonical Higgs bridge. Every rooted witness and relative attachment is retained. The result establishes simultaneous incidence and response laws; it does not require these laws to be instantiated by a common solution of the field equations. A represented local law then exhibits the gauge–matter and charged Yukawa operations explicitly. Its realization domain is distinct from that of the structural incidence map, a distinction that is essential in the neutral sector.

5.1 Identifying the complete bridge, not its name

Two source results are involved in the identification. The metric-envelope paper admits $C_H = H_{\text{br}/\text{fix}}$ as the bridge/fixation carrier, distinct from a scale value or scalar coordinate [12, Def. 6.1 and Thm. 6.2]. The Higgs carrier paper identifies that occupant with the quotient-reduced bridge-witness class, its response signature, reduced mediation and role-rank [13, Def. 7.3 and Thms. 7.4, 7.6]. This is the canonical tuple of MGN3 [14, Defs. 6.2, 7.1 and Thm. 7.3]. These are distinct source statements; the first does not identify every marked response of the second.

Definition 5.1 (Canonical shared-bridge branch). The canonical branch receives the actual bridge source

$$\mathcal{B}_H = (W_{\text{br}}/\sim_{\text{br}}, \Sigma_{\text{br}}, \beta_{\text{red}}, \rho_{\text{br}})$$

with its inherited gauge, order, deformation, residual, guard and continuation data. A fermionic or bosonic bridge port is received through its source map to this full intrinsic tuple. Its presentation comparison is retained together with every incident map. The fermionic ports are those in the actual accepted rooted incidence diagrams; the bosonic ports are those of an actual admitted gauge-Higgs source chart. No identification of their additional arguments is part of the common bridge map.

The full tuple is essential even when a presentation abbreviates it to C_H . Two order inputs may have the same reduced mediation output without being the same bridge witness. Likewise, carrier identity does not specify a chosen vacuum, response value, or coupling tensor.

Proposition 5.2 (Canonical bridge reception). *On the canonical branch, the received fermionic and bosonic bridge ports have one intrinsic source value $\mathcal{B}_H$. Their presentation identifications commute with the bridge-witness, order, reduced-mediation and guard maps. The identification neither identifies the fermionic and electroweak response maps nor identifies distinct marked uses of the bridge.*

Proof. The accepted fermionic incidence family factors through the canonical $H_{\text{br}/\text{fix}}$, with its kind and rooted incidence retained [2, Thms. 5.3–5.4]. The Higgs carrier closure theorem identifies the occupant denoted by that symbol with the full quotient-reduced MGN3 tuple, rather than with a scalar representative [13, Thm. 7.4]. The bosonic reception already retains this same tuple and its incident source data [3, Sec. 11]. Receiving both uses at that canonical source therefore gives identity on the intrinsic tuple; displayed presentation maps are its source-proved comparisons.

For such a comparison the witness class and the order class are transported independently. The reduced map is the same map on those transported classes, so its domain and value agree. This does not infer equality of witness classes from equality of its outputs. MGN3 presentation invariance and bridge rigidity preserve the remaining tuple fields and their inherited quotient [14, Thms. 17.4, 18.1–18.4]. The admitted source comparison also retains the guard and continuation records. Thus all incident squares commute. The extra fermionic and bosonic operand sorts are unchanged and are not identified by this comparison; nor is a particular marked use identified with the whole bridge law. $\square$

The identification is determined by the canonical source maps. A different bridge signature or a relative attachment lacking source admission does not enter this branch merely because it is assigned the same carrier name.

5.2 Reception of the rooted incidence operations

Write F_f for the complete FCGS kind- f rooted diagram. In addition to E_f, η_f, χ_f , it retains the map

$$\beta_{F,f} : Q_f^Y \longrightarrow \mathcal{H}_{F,f}(C_H)$$

and the source, boundary, continuation and failure data of that incidence. Here $\mathcal{H}_{F,f}(C_H)$ is the groupoid of rooted incidence records whose bridge leg is C_H . The map attaches the record to its source-derived response slot. It is not the inverse of a many-input Higgs-response graph.

Theorem 5.3 (Relative reception of fermionic incidence vertices). *Each accepted fermionic incidence vertex is an admitted primitive source operation in the enlarged joint diagram calculus. For a lawful same-kind source comparison $u : X \rightarrow X'$, let u_E, u_S, u_Q, u_H denote its induced maps on positioned roots, rooted components, response slots and incidence records. Then*

$$u_S \eta_{f,X} = \eta_{f,X'} u_E, \quad (1)$$

$$u_Q \chi_{f,X} = \chi_{f,X'} u_S, \quad (2)$$

$$u_H \beta_{F,f,X} = \beta_{F,f,X'} u_Q. \quad (3)$$

The source comparison preserves and reflects the admitted domain of every received restriction and continuation of these vertices, including its exact failure value. It remains valid when the vertex is attached to fixed intrinsic sibling inputs through the shared bridge, provided all its incident maps are transported. No full-target cross-kind comparison is introduced.

Proof. The vertices are the actual maps of the accepted source diagrams, not maps inferred from their component counts. Equations (1)–(2) are the accepted same-target naturality squares; equation (3) is rooted coupling naturality [2, Thm. 3.2 and Thm. 5.4]. The full source fingerprint includes its boundary, continuation, provenance and failure rows, and the inverse comparison must preserve them as well [2, Def. 2.1]. Therefore a restricted operation is admitted before transport if and only if it is admitted afterward. Its value, or its retained failed-law record, is transported by the same source map.

To check a fixed sibling explicitly, use the orientation $v : X' \rightarrow X$ for the changed input presentation. An outgoing incident map m becomes mv , an incoming map a becomes $v^{-1}a$, and an incident endomorphism L becomes $v^{-1}Lv$. Every incident square therefore commutes. The sibling's intrinsic object and the actual relative incidence are unchanged. At the Higgs leg Proposition 5.2 supplies the comparison of the full bridge tuple, including its independent witness and order classes. The same reduced mediation and guard predicates are evaluated on those same intrinsic arguments. Reversing v proves domain reflection.

For a source-licensed compression, use its common intrinsic record and its certified relative descent of incident maps, rather than an inverse of a noninvertible raw map. This is exactly the fixed-input distinction proved for bosonic vertices [3, Thms. 11.4–11.5]. The argument above supplies the additional FCGS vertices individually; their admission has not been inferred merely from membership in the older bosonic network grammar. Finally the source maps preserve f , so the empty full-target cross-kind Hom sets remain empty. $\square$

5.3 A finite common-source construction

Let B be an actual bosonic source network on the canonical branch, including its structural support and vacuum-compatibility record. Attach the four F_f by their shared full bridge maps. When the inputs use different presentations, keep the maps between those presentations. The resulting object, denoted $N_H(F, B)$, is a finite diagram of source charts. It is not defined as a Cartesian product of marginal signature classes.

Theorem 5.4 (Rooted structural common-source assembly). *The diagram $N_H(F, B)$ has a well-defined intrinsic source evaluation and contains the four complete rooted fermionic incidence diagrams and the bosonic structural-support diagram simultaneously. There are inclusion and deletion maps*

$$F_f \xrightarrow{i_f} N_H(F, B) \xrightarrow{d_f} F_f, \quad d_f i_f = 1_{F_f}, \quad B \xrightarrow{i_B} N_H(F, B) \xrightarrow{d_B} B, \quad d_B i_B = 1_B,$$

where deletion means restriction to the indicated marked subdiagram. The inherited rooted occurrences remain distinct and retain their exact accepted incidence maps. Thus the canonical source-network domain is nonempty wherever its received source inputs are present. No universal pointed amalgamation or simultaneous physical terminal occupation is asserted.

Proof. Start with the finite disjoint union of the actual source diagrams and join their designated bridge ports using the maps of Proposition 5.2. This is composition of already given interface maps. It assigns no new output to an old operation and equates no additional kind, root, parameter, boundary or authority field. Every fermionic vertex is source-admitted by Theorem 5.3. Every bosonic vertex is source-admitted by its existing source-network construction [3, Def. 13.5 and Thm. 13.9]. At a joined port the intrinsic input is literally the shared source restriction; the presentation comparison is retained as part of its execution. Successive evaluation of the finite operations therefore defines an intrinsic evaluation on the joined diagram. No equality of separately computed signatures is used to license a vertex.

A lawful presentation change transports each source vertex and its incident maps. The domain reflection laws preserve each successful operation and each failed operation at its original type. Induction over the finite diagram proves presentation independence of its full source evaluation. This combines the new fermionic reception with bosonic functorial evaluation on relative charts [3, Prop. 13.7].

The construction has adjoined vertices and interface maps without altering either constituent diagram. Restriction to a marked constituent therefore gives the displayed retractions. In particular it recovers the literal $E_f, \eta_f, \chi_f, \beta_{F,f}$ and their source witnesses. A new identification of two rooted occurrences would restrict to an identification forbidden by those retained coordinates. No such relation is generated by joining bridge ports, and the external S_3 action is not an orbit quotient of the occurrence object.

The complete accepted fermionic source supplies the four F_f , while the canonical bosonic source supplies the stated B ; their canonical bridge reception supplies the actual joining maps. Hence this gives an object of the source-network domain, not just a list of necessary compatibility tests. The full-source bosonic reception retains the exact antecedents of its source-realized fibre [3, Secs. 12–13]. For a different proposed source without those inputs, this proof supplies no additional existence claim. Finally, the construction is a diagram of actual charts. It requires neither an isomorphism from their formal fibre product to a universal physical record nor a term of a missing terminal-certificate type. $\square$

The simultaneous occupancy is that of structural incidence positions. Nonzero field solutions, pole residues and numerical coupling values require their own realization maps. Although the construction retains several charts, it is more than a collection of marginal records: the bridge maps, source vertices, relative incidences and evaluations are all part of one admitted diagram.

5.4 An explicitly evaluated mixed law chart

The represented branch below exhibits mixed compatibility while preserving the independent construction of the roots. Use the source-derived $p_f : E_f \rightarrow I_3$ to index a multiplicity space $V_I = \mathbb{C}[I_3]$. This is downstream linear notation for the established occurrence set. No matrix dimension enters its formation.

On the minimal chiral representation branch, use integer hypercharge $q = 6Y$ and one of the source-admitted groups

$$G_n = (SU(3) \times SU(2) \times U(1)_q)/\Gamma_n, \quad n \in \{1, 2, 3, 6\}.$$

For each $i \in I_3$, the left-handed representation spaces and the one Higgs space are

$$\begin{aligned} Q_i &= (\mathbf{3}, \mathbf{2})_1, & u_i^c &= (\bar{\mathbf{3}}, \mathbf{1})_{-4}, & d_i^c &= (\bar{\mathbf{3}}, \mathbf{1})_2, \\ L_i &= (\mathbf{1}, \mathbf{2})_{-3}, & e_i^c &= (\mathbf{1}, \mathbf{1})_6, & H &= (\mathbf{1}, \mathbf{2})_3. \end{aligned} \quad (4)$$

Charge and global-form admission follows from the joint derivation, not from the color adjoint [5, Sec. 6][6, Prop. 5.1]. The upper and lower weak components of Q_i have the marked U, D root incidences and those of L_i have the marked N, L incidences. These attachments use the canonical root-position spans and their kind tags. They are not full-target equivalences between the two components. The conjugate singlets retain their left-handed convention maps and source ancestry.

Definition 5.5 (Marked polynomial-law source). At a licensed local representation chart fix one metric and spin convention, one connection $A = (A_3, A_2, B)$, the admitted gauge and Higgs kinetic conventions, and one Higgs projection of the canonical bridge source. Let $\mathcal{P}$ be the graded polynomial algebra of its finite field jets: the bosonic jet coordinates are even and the fermion and conjugate fermion jet coordinates are independent odd generators. Tensor evaluation, gauge action, covariant differentiation, conjugation and contraction are their actual represented operations. An operation is admitted at this polynomial target by a specified multilinear map or derivation with its source representation and covariance witnesses. It is not admitted by assigning it an independently chosen output.

The original rooted source diagrams and bridge incidences remain attached to the chart. The polynomial evaluation is a represented operation on those marked slots, not a definition of their structural response quotient. Any additional standing claim about a Yukawa slot retains its independent coupling-witness and partial-realization domain.

In this chart, with Weyl spinor contraction understood, the charged Yukawa trilinear maps are

$$\begin{aligned} \mathcal{Y}_u &= \sum_{i,j} Y_{ij}^u \epsilon_{ab} Q_i^{\alpha a} H^b u_{j,\alpha}^c, \\ \mathcal{Y}_d &= \sum_{i,j} Y_{ij}^d Q_i^{\alpha a} (H^\dagger)_a d_{j,\alpha}^c, \\ \mathcal{Y}_e &= \sum_{i,j} Y_{ij}^e L_i^a (H^\dagger)_a e_j^c. \end{aligned} \quad (5)$$

Their coefficient tensors are retained law parameters. In particular, changing a coefficient without its source comparison is not a redescription of the same mixed operation.

Theorem 5.6 (Nonzero mixed operations on one represented source). *The representation branch (4) admits a marked polynomial-law chart with the following simultaneous properties.*

1. *All matter and Higgs representations and their mixed operations descend to the same G_n . The inherited local gauge and mixed gravitational anomaly sums vanish, with even weak-doublet parity, for every marked chiral copy.*
2. *Gauge action and the gauge-matter current maps are nonzero on each represented matter root family. The three trilinear map families in (5) are nonzero invariant maps. Choosing nonzero coefficients gives actual nonzero polynomial operations, not merely displayed candidate outputs.*

3. For $\lambda > 0$, $v > 0$, the same Higgs field has the admitted scalar potential $V(H) = \lambda(H^\dagger H - v^2/2)^2$ with stationary minimum $h_0 = (0, v/\sqrt{2})^\top$. Its residual group is the color-correlated stabilizer of the same global gauge group, with the inherited gauge support ranks $(8, 1, 2, 1)$.
4. Whole-diagram source transport preserves these operation graphs. At a fixed intrinsic sibling, a changed input presentation together with its incident coefficient maps preserves and reflects the polynomial operation domain and transports its value.

These conclusions concern a populated operation-law chart. They do not assert nonzero classical fermion solutions, a quantum state, a numerical Yukawa prediction, or full MGN4 standing for every displayed matrix entry.

Proof. For a center label (t, s, q) , descent is exactly $2t + 3s + q = 0 \pmod{n}$. Every representation in (4) satisfies the stronger congruence modulo six. The per-copy anomaly calculation is the full joint calculation in [6, Prop. 5.1]; summing over the actual already formed index set preserves its zero values. This verifies the stated anomaly data, not an unregistered global-anomaly conclusion.

The color pairings in (5) are fundamental–antifundamental evaluation. The weak maps are its invariant alternating form or the doublet–dual evaluation. Their hypercharge sums are respectively

$$1 + 3 - 4 = 0, \quad 1 - 3 + 2 = 0, \quad -3 - 3 + 6 = 0.$$

Thus they are actual intertwiners for the one represented group, including its central quotient. Choose paired nonzero color and Weyl basis vectors; for the first weak contraction choose the upper Q component and the lower h_0 component, and for the other two choose the lower doublet component. The resulting coefficients of products of distinct odd generators are nonzero. Hence the trilinear maps themselves are not zero. The finite sum is nonzero whenever a selected coefficient is nonzero, as different indexed monomials are linearly independent before any additional field equation is imposed.

For the current statement choose an internal generator whose represented matrix has a nonzero entry in the indicated matter representation. Such a generator exists in each of the displayed nontrivial representations. The corresponding coefficient in $\bar{\psi}\bar{\sigma}^\mu d\rho(X)\psi$ is then a nonzero product of independent odd generators. Color singlets use their weak or hypercharge action, and the neutral weak component uses its full weak/hypercharge action rather than an electromagnetic charge. Gauge covariance follows from the representation law and invariant evaluation. The covariant derivative uses the same A and the same coefficient convention at every vertex.

The Higgs potential is nonnegative and vanishes at h_0 ; direct differentiation gives zero first derivative there. The representation, kinetic Gram form and nonzero Higgs projection are those of the admitted local electroweak–Higgs branch [4, Secs. 9–10]. Its stabilizer in G_n and its gauge support decomposition are the actual joint results [6, Thm. 5.3 and Sec. 6]. Neither the group nor the Higgs field is chosen separately for a different sector. In the odd algebra the potential minimum is a bosonic fixation record; it is not a statement that every possible odd configuration solves the coupled equations.

Finally, consider a multilinear operation M and invertible input and output presentation maps $U_1, \dots, U_k, U_o$. Its transported tensor is

$$M' = U_o M (U_1^{-1} \otimes \dots \otimes U_k^{-1}).$$

Therefore

$$M'(U_1 x_1, \dots, U_k x_k) = U_o M(x_1, \dots, x_k).$$

For a fixed sibling take its U_j to be the identity. Invertibility proves the converse domain implication. Polynomial operations are defined on their stated jet sorts; source restrictions retain any additional domain predicates through Theorem 5.3 and the bosonic source reflection

theorem. Derivations obey the same intertwining relation, including the connection transformation. Induction gives the claim for every finite admitted network. A changed shared Higgs presentation transports all its incident tensors, not one selected Higgs occurrence alone. $\square$

The operation witnesses in this proof are the actual representations, invariant tensors, derivations and their covariance equations. They establish the declared polynomial-law admission. They do not replace a stronger order-deformation coupling witness.

Definition 5.7 (Standing coupling realization domain). Let D_Y consist of the actual source incidences that supply an admitted MGN4 coupling witness together with its represented map into the polynomial-law chart. The record retains its canonical bridge factorization, product, representation, charge, anomaly, residual, order and deformation maps, its coupling quotient, and any requested scale transport. The evaluation

$$\text{Realy} : D_Y \longrightarrow \text{Op}(\mathcal{P})$$

is the represented operation of that witness.

Proposition 5.8 (Exact coupling authority boundary). *On D_Y , coupling realization preserves the original structural incidence and its admitted source comparisons. The image of Realy need not be all the invariant operations in (5). In particular, gauge invariance, a nonzero matrix entry, or a stationary Higgs value alone does not supply the other fields of D_Y .*

Proof. The source coupling witness is formed before its coordinate presentation. Its actual factorization through the bridge and inherited ledgers is the admission condition of MGN4 [15, Def. 5.3, Secs. 7–12 and Thm. 26.1]. Its represented map is an incidence map in the joint source diagram. Transport of that map commutes with evaluation by the relative source theorem, proving preservation on its domain. The polynomial invariance calculation checks representation and charge compatibility and the stated anomaly data. It does not construct a missing order/deformation witness, scale-transport record or standing nonzero certificate. Thus no surjectivity onto all invariant matrix presentations follows. $\square$

5.5 Neutral incidences and stronger realization layers

The neutral structural map $\beta_{F,N} : Q_N^Y \rightarrow \mathcal{H}_{F,N}(C_H)$ is present in the common-source theorem. The minimal chiral branch of (4), however, has no right-handed neutrino. It therefore does not contain a renormalizable Dirac neutrino Yukawa vertex. A heavy-neutral completion or a higher-dimensional operator requires its own source and target reception; neither is inserted to turn the neutral structural incidence into another member of (5).

Corollary 5.9 (Separation of structural and realized population). *The common-source network contains all the accepted rooted fermionic incidences through one Higgs carrier, while its represented law can retain a partial coupling realization map. BCRS-III's occupied classical branch supplies a separate simultaneous gauge-Higgs-gravity-scalar history with zero fermion fields. Neither construction alters the frozen FCGS terminal-apex result or fills an unreceived quantum, pole or confinement realization domain.*

Proof. Structural population follows from Theorem 5.4; the partial coupling boundary is Proposition 5.8. The classical history and its source reception are exact predecessor results [6, Def. 7.1 and Thms. 7.7, 7.9]. Their zero fermion coordinates do not remove the carrier, but they are not nonzero fermion solutions. The positive terminal witness types required by the frozen fermionic apex are different from rooted incidence and polynomial operation types [2, Sec. 8 and Thm. 10.11]. No map constructed here has those missing terminal types as its codomain. The stronger physical domains likewise retain their separate source constructors. $\square$

5.6 Reconstruction on the finite-network source image

Proposition 5.10 (Complete-diagram reconstruction). *For the constructed finite-network branch, retain every constituent rooted and bosonic chart, every actual relative map, and every mixed operation tensor and domain predicate that defines the network. Isomorphism of these complete records is exactly lawful isomorphism of the joint diagram. Therefore their record classes reconstruct the source classes on that branch. Forgetting a mixed tensor or relative map can destroy this property even when the marginal classes are unchanged.*

Proof. A joint-diagram isomorphism restricts to isomorphisms of the displayed records. Conversely a comparison of the complete records provides its map at every vertex and commutes with every incident arrow and admission predicate. These are all the defining fields of this branch, so the maps assemble into a lawful diagram isomorphism. This establishes comparison reflection, the missing implication in a mere object-coverage argument [6, Thm. 4.2].

For the final claim fix two one-dimensional operand spaces and compare the bilinear maps $M(x, y) = xy$ and $M'(x, y) = 2xy$ with the operand marks and comparisons fixed. Their marginal source records agree, while evaluation at $(1, 1)$ separates the mixed operations. Removing the coefficient tensor would erase that difference. The example changes neither root cardinality nor Higgs carrier identity; it changes retained joint response content. $\square$

6 A canonical common source for the structural atlas

The finite-network result does not imply that independently chosen preparations have a common extension. At structural-law resolution, one coherent Standard-Model source provides a stronger construction: the matter and bosonic records are obtained as its simultaneous restrictions. The source supplies their common compatibility data, while the neutral root proof remains independent and physical projection remains downstream.

6.1 The coherent structural source

The standalone structural spine supplies the witness-bearing object

$$\Sigma_{SM} = (G_{SM}, R_{SM}, H_{\text{br/fix}}, O_Y, Q_{SM}, A_{\text{anom}}, V_{SM}, M_{\text{slot}}, \Theta_{\text{mix}}, N_g) / \simeq_{\text{Adm}}$$

[19, Def. 1.7, Sec. 9.3 and Thm. 9.7]. This source is more than its displayed list of entries. Its gauge, representation, charge, anomaly, vacuum, bridge and response entries have the common source witnesses of the structural spine. The received global coherence theorem says that these form one carrier architecture [8, Sec. 7 and Thm. 7.1]. In particular, the charge and anomaly records concern the same represented product, and the vacuum-supported residual record concerns that product and representation ledger. A tuple of separately surviving projections would not have this property.

At the earlier gauge-skeleton stage, the common witness ledger retains boundary, transport, loop residue, representation, charge, anomaly, residual, vacuum and deformation fields [20, Def. 5.1 and Sec. 5.1]. The Higgs and coupling stages add their actual mediation and compatibility witnesses over that ledger, not over independently selected copies of it [19, Defs. 1.9 and 1.12, Thms. 9.3–9.4]. Here V_{SM} is the received vacuum-support and residual-compatibility structure. It is not a symbol meaning that any proposed vacuum is admitted.

The coupling entry has the exact type

$$O_Y = (O_{Y,u}, O_{Y,d}, O_{Y,e}; O_{\nu, \text{status}}).$$

Its charged entries record operator availability and coupling-witness topology. Its neutral entry retains the declared operator-status domain. Absent and obstructed neutral operators are not replaced by a Dirac operator. Likewise Θ_{mix} retains whatever relative, transition and CP-residue witnesses the source actually supplies, with their partial domains. It is neither a freely chosen matrix nor an assertion that every mixing slot is populated [19, Defs. 1.7 and 1.12] [16, Defs. 3.3–3.5 and 6.1–6.2].

Definition 6.1 (Canonical structural-law branch). The canonical branch receives an actual witness-bearing presentation s of the structural spine output, together with its source comparisons. It retains the full structural source laws, their operator-status and failure domains, the common bridge tuple, the joint representation–charge–anomaly–vacuum incidences, and the source boundary and continuation interfaces. Every current sector record is a restriction of this one presentation.

The complete retained demand is structural-law agreement. A selected numerical order parameter, a pointed vacuum preparation, a chosen physical history, a pole, a confinement state or a terminal matter certificate is not a field of this demand. If such a field is requested, its independently established domain is added to the target before the corresponding population question is asked. In particular, retaining a continuation law does not imply that every requested history is defined.

Existence of this branch follows from the structural derivation and its common-ledger witnesses, at the stated realized-interior envelope [19, Thm. 9.7]. The earlier bookkeeping existence result [11, Lem. 6.12] is consistent with this result, but does not replace the stronger source-spine witnesses. The conditional assembly criterion of SR7 [20, Prop. 16.1] likewise does not assert that arbitrary marginal survivor records have an extension.

6.2 Reindexing the whole source over the derived roots

The count-free matter cell of FCGS–I is extracted from this same structural source. Its generation-erasure map deletes the source family number, names, indices, family-indexed matrix dimensions and numerical coordinates. It retains representation, color, weak component, relative charge, chirality, bridge identity, operator-status, boundary and continuation types [1, Sec. 5.1]. Write its four composite roles as

$$\mathfrak{F}^{\text{cf}} = \{r_N, r_L, r_U, r_D\}.$$

Only after the primitive generation architecture and its complete neutral occupation have been received does the source-derived base change form $I_3 \times \mathfrak{F}^{\text{cf}}$ [1, Defs. 6.1–6.2 and Thm. 6.3].

For the present joint construction it is important not to erase genuine intergeneration incidences along with their names. Let J_s denote the native family alphabet of the received source presentation. Its family envelope is part of that source. It is received with complete primitive-root occupation, rather than used to derive I_3 . The exact endpoint [21, Theorems 5.4 and 6.2] and the FCGS root receipt [1, Sections 3–4] then identify this native alphabet with the whole three-position base. Form the entire marking set

$$\mathfrak{M}_s = \text{Bij}(I_3, J_s).$$

It is nonempty because both sets have three elements. It is an external marking torsor, not a selected numbering of the derived roots.

For $m \in \mathfrak{M}_s$, pull back *every* family-indexed field of s along m . A relation on J_s^k pulls back along m^k ; an indexed map has all input and output legs transported; a guard, boundary attachment or failure predicate is precomposed with the same transport. Denote the whole reindexed presentation by $s[m]$. This operation does not change the gauge, Higgs or vacuum source. It changes only the alphabet used to display its family incidences.

Lemma 6.2 (Whole-source reindexing). *Each $s[m]$ is a source presentation with exactly the same intrinsic structural laws, domains and compatibility witnesses as s . Every family-indexed incidence, including a nontrivial relative mixing or CP witness, is retained. Different markings are related by simultaneous source re-presentation, not by an assertion that every permutation stabilizes a fixed displayed tensor.*

Proof. Let T_m denote the bijection on a relevant indexed argument space. For a partial source map $g : D_g \rightsquigarrow Y$, its reindexed graph is

$$g_m = T_m^{-1} g T_m, \quad D_{g_m} = T_m^{-1}(D_g),$$

with separately typed transports on different input and output spaces. For a many-input map this formula uses the product of its input transports. Applying T_m to a source equality proves the transported equality; applying the inverse proves reflection. The same argument applies to each compatibility, standing, boundary and failure predicate. Identity and composition are preserved by cancellation of adjacent inverse transports. Thus the full source diagram, not just its marginal matrices, is re-presented.

For two markings, $m'^{-1}m$ gives the external change of root alphabet with every other indexed coordinate transported simultaneously. In the assembled atlas this also transports the retained ancestral root maps and records by their accepted external S_3 action. Holding an ancestral mark or a relative attachment fixed instead restricts the comparison to its actual stabilizer; changing m alone need not be skin. A permutation fixes a displayed tensor only if it belongs to that tensor's actual stabilizer. If a more restrictive target fixes boundary names or apparatus marks, only comparisons preserving those additional data remain; it is a different pointed comparison problem. $\square$

This proof uses $|I_3| = 3$ only after the independent primitive root-occupation theorem. It obtains neither a second root count nor a native quark holonomy rank from the source family alphabet. Up and down kinds receive the same root base and retain different weak, charge and Higgs-operator incidences.

Primitive generation receipt. *The structural source above receives exactly three primitive generation positions, independently of its choice of marking. This root count does not count the four matter kinds, bosonic whole-law roles, local response dimensions or terminal bearers.*

Proof. The primitive-standing theorem fixes the architecture at three, and the FCGS complete occupation identifies its neutral occurrence base [21, Theorems 6.2 and 5.4][1, Sections 3–4]. Every member of $\mathfrak{M}_s$ transports the entire received occupation to J_s . The whole-source reindexing lemma preserves the actual indexed maps, joint domains, guards and failure records, rather than only their cardinalities. The count is therefore constant on the full marking family [21, Theorem 6.6]; no preferred labeling or tensor stabilizer is inferred. Each of the other inventories is the image or domain of its separately typed source map. $\square$

6.3 The rooted restriction and its ancestral comparison

Fix $s[m]$. Its count-free matter restriction is the same $\mathfrak{F}^{\text{cf}}$ used in FCGS-I: neutral and charged lepton-pair members, and up and down quark-pair members, with the complete relative representation and operator-status coordinates. Position these roles over I_3 . The source incidence at (i, r_f) is the kind- f restriction of s at $m(i)$, not an occupant selected by mass or matrix entry. Write

$$\rho_f^{s,m} : \text{Disc}(I_3 \times \{r_f\}) \longrightarrow \mathcal{I}_f(s[m])$$

for this rooted source-incidence map. Its codomain retains the source kind, operator-status domain, common Higgs bridge and all current incidences needed by the canonical structural-law demand.

The existence here is the existence of the source role at each source family position, received in R_{SM} and O_Y . Explicitly, restrict the actual composite matter role and all its source incidence fields at $m(i)$, and retain the root i . This constructs $\rho_f^{s,m}$ without asserting that every hypothesis of a new complete FCGS response target has been activated. The FCGS positioned occupancy and rooted-realization theorems continue to apply to their fixed R_f, C_f [1, Thms. 6.5–6.6]; they are not transferred to a new full target by equality of a count-free reduct.

For a neutral absent or obstructed operator, the source incidence retains that typed structural status. It is not a positive coupling. This is consistent with the exact status discipline of the FCGS response arena [1, Def. 7.1]. No assertion is made here about the component quotient of all possible new full response objects. In particular, no new quotient is declared to have one component merely because a root label was appended to a record.

To compare this current source with an earlier complete FCGS record, specify the common reduct explicitly. It retains I_3 , the positioned kind, the count-free representation/color/weak/relative-charge/chirality coordinates, the full common bridge identity and the source operator-status, boundary and continuation *types*. In particular, an operator-status law is not replaced by a selected value of that law. Let $\mathcal{R}_f^{\text{cf}}$ be the groupoid of those rooted cell records. The ancestral and current incidence records have restriction maps d_f^0 and $d_f^{s,m}$ to this explicit reduct. The inherited source comparison induces a specified natural isomorphism

$$d_f^{s,m} \rho_f^{s,m} \cong_{\vartheta_f^{s,m}} d_f^0 \beta_{F,f} \chi_f \eta_f : \text{Disc}(E_f) \longrightarrow \mathcal{R}_f^{\text{cf}}, \quad E_f = I_3 \times \{r_f\}. \quad (6)$$

Lemma 6.3 (Source-reduct comparison of the rooted chains). *The source-reduct square (6) commutes through the specified source comparison. It preserves the root and all displayed count-free cell fields. It does not assert an arrow between incompatible complete FCGS target fingerprints.*

Proof. On (i, r_f) , the accepted composite attaches the root i to its source-derived role and bridge slot. The current composite restricts the same source-derived count-free role at $m(i)$ and pulls its alphabet back to i . The role coordinates agree because FCGS–I extracted them by the very generation-erasure projection of Σ_{SM} received here [1, Sec. 5.1] [19, Def. 1.7]. The bridge coordinates agree by the full carrier reception proved above, and all operator-status and interface types are those retained by that projection. Restricting the actual comparison of the two presentations of that source gives the component $\vartheta_f^{s,m}(i, r_f)$. The domain is discrete, so these components form the stated natural isomorphism; simultaneous root transport respects external marking. No source representative is selected to turn an isomorphism into literal equality.

The earlier provenance, realized boundary, continuation history, complete response status and failure records remain attached to the earlier record. They are not entries of the displayed common reduct and are not claimed equal to different current source arguments. The comparison is therefore a source-reduct comparison, not an application of full-target naturality outside its lawful domain [2, Defs. 2.1 and 3.3, Thm. 3.2]. $\square$

6.4 One source, all structural compatibility fields

Let $B(s[m])$ be the bosonic source restriction: its gauge-product and response laws, common Higgs mediation, admitted order/residual/vacuum-support relations and their actual domains. This is a source-law record in the BCRS reception theory, not necessarily one of its independently populated physical projections. It is evaluated using the received source constructors and their exact domain reflection [6, Secs. 4–5]. The common structural fields are specified below, before the observational quotient is formed.

Shared field	Construction from the one source
Product, representation and charge	Restrict the same G_{SM}, R_{SM}, Q_{SM} ; duals, weak pairing and color incidences use their actual source maps.
Joint anomaly	Retain the one anomaly ledger on those same representations and charges; do not combine unrelated statements that a marginal anomaly vanishes.
Higgs and order/deformation	Restrict the one full $H_{br/fix}$ tuple, its admitted order/deformation witness maps, and their domains.
Vacuum and residual support	Restrict the actual V_{SM} support and compatibility relations over the same product, anomaly and bridge source.
Fermionic response	Use the source O_Y topology and status domain at each positioned role; retain the neutral status rather than invent a neutral polynomial interaction.
Relative mixing and CP	Transport every actual Θ_{mix} witness and all of its attachments together; no independent matrix is substituted.
Boundary, continuation, failure	Use the common source interfaces and partial-operation domains. Pointed histories require their own admitted arguments and are not inferred from a type.

Theorem 6.4 (Canonical structural single-source apex). *On the received canonical structural-law branch, the same source presentation $s[m]$ supports all four current rooted matter restrictions $\rho_f^{s,m}$, the integrated bosonic restriction $B(s[m])$, and every shared structural field in the preceding ledger. These restrictions form an actual common-source diagram. Their simultaneous compatibility is witnessed by their maps from $s[m]$, not imposed after observing their marginal response values.*

The branch is populated by receiving the structural-spine source and retaining its full nonempty marking torsor. Every marking gives the same intrinsic diagram up to whole-source re-presentation. Its rooted restriction agrees with the accepted FCGS chain at the explicitly retained source reduct (6); all earlier complete records remain unchanged ancestors. The conclusion is a simultaneous structural-law apex. A pointed vacuum, physical continuation or terminal apex is a separate extension of this target.

Proof. Receive the actual structural-spine output with the common-ledger witnesses specified above. This is the source existence input, supplied before the CSP-S quotient and before any selection of sector responses. The independently proved I_3 and the source family envelope give the nonempty marking torsor. Whole-source reindexing preserves every source map, guard and identity by Lemma 6.2.

Take the displayed restrictions of $s[m]$. For any shared source field a , its restriction through the fermionic diagram and its restriction through the bosonic diagram are both the same component map $a(s[m])$. They therefore agree with the same attachments. In particular, the anomaly and vacuum predicates are evaluated on the same product and representation inputs, while Higgs mediation uses the same order/deformation domain. The source coherence witnesses prove all these predicates simultaneously. This is stronger than agreement of a short list of carrier names or of separately computed signatures.

Each current role is the already existing source role at its reindexed position. Restriction of its actual incidence fields constructs the rooted map without a new full-response quotient. The source-reduct comparison is Lemma 6.3. The bosonic restriction uses the same source constructors and introduces no independent vacuum or Higgs argument. Hence the complete diagram has one actual source at the declared structural-law resolution.

Finally retain, rather than select, all markings. The transport between two markings carries every relative incidence, so the intrinsic diagram is unchanged. None of the construction uses the missing physical fields listed in Definition 6.1. It therefore neither supplies those fields nor changes an earlier terminal disposition. $\square$

The common-source restriction is the essential step. Arbitrary pointed Higgs or matter records

need not possess a joint extension; the records considered here are compatible because they restrict one coherent source. Their full-target ancestors remain distinct from these current structural restrictions. Additional physical arguments require their own exact source admission.

7 Reconstructing the mixed interaction law

The common source fixes the objects and incidences to be compared. A represented interaction law also carries relative information that separate spectra may omit. Reconstruction therefore requires probes of the joint law. The following results make this distinction exact without presupposing numerical values for any coupling.

7.1 The independently specified coefficient domain

Fix the marked polynomial-law chart of Definition 5.5. Its representation, spin, metric, kinetic-normalization and source-attachment records are held fixed. The flavor modules are finite-dimensional Hermitian spaces supplied by this representation; the established occurrence set supplies their marking. The linear notation is a realization of the root positions and does not generate the roots. In particular the existence of three source occurrences alone does not supply a mass operator, a Hermitian structure or a propagation law.

The coefficient target has the three charged trilinear maps $\mathcal{Y}_u, \mathcal{Y}_d, \mathcal{Y}_e$, their conjugates, and the gauge, Higgs and mixed operations of the displayed local law. There is no neutral Dirac coefficient in this target. When the target requires standing mass or flavor transport, its domain is the image of actual coupling and common-interface witnesses, not the space of all matrices. This is the distinction between an invariant coupling slot and an admitted standing coupling in MGN4 [15, Def. 5.3 and Thm. 26.1], and between a displayed mixing matrix and an independently standing common interface in MGN6 [16, Thms. 5.3, 5.5 and 13.2].

Definition 7.1 (Coefficient observation target). The marked coefficient target observes the multilinear operation tensors themselves. Its elementary probes are their components in the fixed tensor bases, equivalently evaluation on basis arguments followed by a coordinate functional on the output. For a graded polynomial these are coefficient-extraction functionals on distinct ordered monomials. They do not substitute ordinary real numbers for odd generators. A source-restricted coefficient target retains the actual witness domain, every source comparison and all additional guards.

These component probes resolve the mathematical tensor law; they do not assert that every component is directly measurable in an experiment. No additional source carrier is introduced. A physical-observable target may be coarser and has its own restriction map.

Lemma 7.2 (Exact recovery of a multilinear operation). *Let $V_1, \dots, V_k, W$ be finite-dimensional vector spaces with fixed bases. The coefficients*

$$M_{a_1 \dots a_k}^b = e^b(M(e_{a_1}, \dots, e_{a_k}))$$

determine a multilinear map $M : V_1 \times \dots \times V_k \rightarrow W$ uniquely. For the three charged invariant contractions in (5), finitely many such probes recover every entry of Y_u, Y_d, Y_e . On any actual source-restricted domain this remains a reconstruction on its image, without asserting that arbitrary coefficient lists lie in that image.

Proof. Expand each input in its basis and use multilinearity. The output is the sum of the displayed coefficients multiplied by the input components and the output basis vectors. Thus equal coefficients give the same map on every argument, and a differing coefficient is separated by its basis probe.

For the up contraction choose one paired color index, one pair of distinct Weyl indices and weak indices with $\epsilon_{ab} = 1$. Its ordered odd monomial has coefficient Y_{ij}^u , up to the fixed nonzero convention factor. For the down and charged-lepton contractions choose paired weak indices instead. Different i, j give distinct ordered monomials. Their coefficient functionals therefore recover each entry, dividing only by that fixed nonzero factor. No inverse mass eigenvalue and no assumed nonzero matrix entry is used. Restriction to a source domain changes the attainable coefficient tuples, not the uniqueness of their evaluations. $\square$

Theorem 7.3 (Complete marked local-law reconstruction). *On the stipulated finite polynomial-law chart, retain the independently observed bosonic law data, the three charged coefficient tensors, and the fixed kinetic, representation, derivative, global, boundary and source-attachment records. These data reconstruct every received finite polynomial operation and every composite built from them. Coefficient probes recover the charged tensors. If the remaining bosonic law probes reconstruct their law data, equality of this combined probe family is equivalent to equality of the whole marked law.*

On a restricted standing domain the same equivalence additionally requires equality of its domain and failure observations and of any retained source field not determined by the law. The statement is about the specified law and its actual image; it does not classify a full quantum theory by finitely many classical coefficients.

Proof. The marked law is specified by its fixed operations, the bosonic law tensors and the three charged tensors. Lemma 7.2 recovers the latter. Reconstruct the former using their actual admitted law probes, as in the local bosonic reconstruction [6, Sec. 6 and Thm. 10.6]. Insert the recovered tensors into the fixed sums, contractions and derivations. This reconstructs each primitive operation. Induction over its expression reconstructs every composite. Conversely equality of the whole law includes equality of each component probe.

If admission also depends on a standing, scale, boundary or continuation witness, its value is not recovered from a polynomial merely because both are associated with the same carrier. Retaining those exact source fields and their failure tests supplies the restricted domain. The same induction then works up to the first failed operation, where the retained diagnostic agrees. No unlisted interaction term, loop completion or boundary sector is claimed to have been reconstructed. $\square$

For example, if $\mathcal{L}_0$ denotes the fixed non-Yukawa part and T_α are the distinct charged monomials with their conjugates, the relevant identity is simply

$$\mathcal{L} = \mathcal{L}_0 + \sum_{\alpha} c_{\alpha} T_{\alpha}.$$

The finite coefficients determine this expression. Adding an independently admitted higher-dimensional operator changes the target and adds its tensor and probes; it is not absorbed into an old coefficient by selecting a convenient representative.

7.2 Finite reconstruction of relative flavor data

The up and down couplings share a left-handed quark module, so their separate singular-value lists need not determine their relative position. A finite family of joint invariants resolves this question, including singular and reducible cases. The coefficient maps are already admitted: the result neither constructs the fermionic roots nor assigns a physical mass or pole to a coefficient.

For an integer $n \geq 1$, let $A, B \in M_n(\mathbb{C})$ be Hermitian. A word is a finite string in two letters, with the empty word evaluated as the identity matrix. Write $w(A, B)$ for its evaluation and $|w|$ for its length. Define

$$\mathcal{I}_n(A, B) := (\text{tr } w(A, B))_{|w| \leq 2n^2 - 1}. \quad (7)$$

The signature retains the complex traces, not only their real parts. The family is sufficient; no minimality claim is required. Its values are considered on the actual image of admitted Hermitian pairs; an arbitrary tuple of numbers need not be the signature of such a pair.

Lemma 7.4 (Finite word span and multiplication reconstruction). *Put $m = n^2 - 1$. The words of length at most m span the unital $*$ -algebra $\mathcal{A}$ generated by A, B . If Hermitian $A', B' \in M_n(\mathbb{C})$ have the same signature (7), there is a unique linear map*

$$T : \mathcal{A} \longrightarrow \mathcal{A}', \quad T(w(A, B)) = w(A', B') \quad (|w| \leq m).$$

It is a unital trace-preserving $$ -algebra isomorphism and satisfies $T(A) = A', T(B) = B'$.*

Proof. Let V_k be the span of words of length at most k . Then

$$V_0 = \mathbb{C}I, \quad V_{k+1} = V_k + AV_k + BV_k.$$

Once $V_{k+1} = V_k$, left multiplication by both generators preserves V_k , so all subsequent word spaces equal it. Since $\dim M_n(\mathbb{C}) = n^2$ and $\dim V_0 = 1$, failure of stabilization by $k = m$ would require at least n^2 strict dimension increases, which is impossible. Thus V_m contains every word. Reversing words shows that it is closed under adjoints, and it is precisely $\mathcal{A}$.

For words u, v of length at most m , their Hilbert–Schmidt Gram entry is

$$\langle u(A, B), v(A, B) \rangle = \text{tr}(u(A, B)^* v(A, B)).$$

Because the generators are Hermitian, this is a trace word of length at most $2m$. Equality of the signatures therefore gives equal Gram matrices. A linear combination of words vanishes exactly when its Hilbert–Schmidt norm is zero, so the two word-evaluation maps have the same linear kernel. Hence T is well-defined, bijective and isometric, with $T(I) = I$.

For $X \in \mathcal{A}$ and every word v of length at most m , trace words of length at most $2m + 1$ give

$$\langle T(v), T(AX) \rangle = \langle v, AX \rangle = \langle T(v), A'T(X) \rangle,$$

where the unprimed evaluations of words are abbreviated in the inner products. Both vectors compared on the right and left belong to $\mathcal{A}'$, which the $T(v)$ span. Therefore $T(AX) = A'T(X)$; likewise $T(BX) = B'T(X)$. This argument also applies when $n = 1$, in which case $m = 0$ and the length-one traces recover the scalar generators.

Induction now gives $T(wX) = w'T(X)$ for every word, without a length restriction. Taking $X = I$ and then extending linearly proves $T(XY) = T(X)T(Y)$. Word reversal and conjugation of coefficients give $T(X^*) = T(X)^*$. Finally $\text{tr } T(X) = \langle I, T(X) \rangle = \langle I, X \rangle = \text{tr } X$. Thus T is the stated isomorphism. $\square$

Lemma 7.5 (Unitary implementation at equal trace). *Every unital trace-preserving $*$ -algebra isomorphism $T : \mathcal{A} \rightarrow \mathcal{A}'$ between unital $*$ -subalgebras of $M_n(\mathbb{C})$ is implemented by a unitary matrix:*

$$T(X) = UXU^* \quad (X \in \mathcal{A}).$$

No irreducibility assumption is required.

Proof. Choose a nonzero projection of smallest rank in $\mathcal{A}$, then repeat in the orthogonal complementary corner. This terminates in mutually orthogonal minimal projections $e_1, \dots, e_r$ whose sum is I . For each i , $e_i \mathcal{A} e_i = \mathbb{C}e_i$: otherwise a nonscalar Hermitian element of this corner would have a nonzero proper spectral projection, polynomial in that element and hence in the corner, contradicting minimality.

Declare $i \sim j$ when $e_i \mathcal{A} e_j \neq 0$. A nonzero x in this corner satisfies $x^*x = \alpha e_j$ and $xx^* = \beta e_i$, with $\alpha, \beta > 0$. Associativity gives $\alpha x = x(x^*x) = (xx^*)x = \beta x$, so $\alpha = \beta$. After rescaling,

this gives a partial isometry with initial projection e_j and final projection e_i . Adjoints and products of these partial isometries show that $\sim$ is an equivalence relation. Each nonzero corner is one-dimensional: if $v^*v = e_j$, $vv^* = e_i$, then $x = v(v^*x)$ and $v^*x \in \mathbb{C}e_j$.

In each equivalence class choose a base index 0, and partial isometries $v_i \in e_i \mathcal{A} e_0$ with $v_i^*v_i = e_0$, $v_i v_i^* = e_i$, taking $v_0 = e_0$. The elements $v_i v_j^*$ are matrix units and span that block; all corners between distinct classes vanish. This constructs the block decomposition directly inside the given matrix algebra.

The images $e'_i = T(e_i)$ and $v'_i = T(v_i)$ have the same relations. Trace preservation gives $\dim e_0 \mathbb{C}^n = \text{tr } e_0 = \text{tr } e'_0 = \dim e'_0 \mathbb{C}^n$. Choose a unitary $U_0 : e_0 \mathbb{C}^n \rightarrow e'_0 \mathbb{C}^n$ for each block and define on its mutually orthogonal summands

$$U(v_i \xi) = v'_i U_0 \xi, \quad \xi \in e_0 \mathbb{C}^n.$$

Each v_i is an isometry on its initial subspace, so these maps assemble to a unitary on $\mathbb{C}^n$. They carry every matrix unit $v_i v_j^*$ to its image under T . Since the matrix units span $\mathcal{A}$, they implement T on the entire algebra. $\square$

Theorem 7.6 (Finite simultaneous Hermitian reconstruction). *For Hermitian pairs of the same size n ,*

$$\mathcal{I}_n(A, B) = \mathcal{I}_n(A', B') \iff \exists U \in \text{U}(n) : \quad A' = UAU^*, \quad B' = UBU^*.$$

Thus traces of words of length at most 17 suffice when $n = 3$. The statement includes repeated eigenvalues, scalar matrices, rank-deficient matrices and reducible pairs.

Proof. Equality of the finite signatures gives the trace-preserving $*$ -isomorphism of Lemma 7.4. Apply Lemma 7.5 to its two generators. Conversely simultaneous conjugation conjugates every evaluated word and preserves its trace. No step of either lemma removes a degenerate or reducible stratum. $\square$

Corollary 7.7 (Two Yukawa maps with a common left module). *Let $Y_u, Y_d, Y'_u, Y'_d \in M_n(\mathbb{C})$ be already admitted coefficient maps between fixed Hermitian flavor modules. Put*

$$A = Y_u Y_u^*, \quad B = Y_d Y_d^*, \quad A' = Y'_u Y'^*_u, \quad B' = Y'_d Y'^*_d.$$

Their signatures $\mathcal{I}_n$ agree if and only if there are $U, V_u, V_d \in \text{U}(n)$ such that

$$Y'_u = U Y_u V_u^*, \quad Y'_d = U Y_d V_d^*.$$

This is classification for the displayed full unitary flavor action, including rank-deficient maps. It is not a claim that every such transformation is a lawful comparison of the complete source.

Proof. The displayed transformation law gives simultaneous conjugation of the left Gram pair, hence equality of signatures. Conversely obtain U from Theorem 7.6. For either coupling put $M = UY$, $N = Y'$. Then $MM^* = NN^*$, and the map

$$M^* \xi \longmapsto N^* \xi$$

is a well-defined isometry from $\text{im } M^*$ onto $\text{im } N^*$, because its inner products are determined by the equal left Gram matrices. Their orthogonal complements have equal dimension, so extend this isometry to a unitary V on the right module. Then $N^* = VM^*$, or $N = MV^*$. Apply this separately to Y_u and Y_d . The extension on the null complements requires no inverse of either coefficient map and proves the rank-deficient case as well. $\square$

Example 7.8 (Equal marginals and different relative data). Take

$$A = \text{diag}(1, 2, 3), \quad B_1 = \text{diag}(1, 2, 4), \quad B_2 = \text{diag}(2, 1, 4).$$

The pairs (A, B_1) and (A, B_2) have identical separate spectra. Nevertheless $\text{tr}(AB_1) = 17$ and $\text{tr}(AB_2) = 16$, so they are not simultaneously unitarily conjugate. All three matrices are positive definite and can be left Gram matrices. The example therefore already occurs on a nonsingular coefficient domain; it is not an artifact of a zero singular value.

Example 7.9 (A complex mixed invariant distinguishes conjugate pairs). Let

$$A = \text{diag}(1, 2, 3), \quad B = \begin{pmatrix} 4 & 1 & -i \\ 1 & 4 & 1 \\ i & 1 & 4 \end{pmatrix}.$$

The matrices B and $\bar{B} = B^\top$ have the same spectrum. They are positive definite, since the off-diagonal contribution to v^*Bv has absolute value at most $2\|v\|^2$, giving $v^*Bv \geq 2\|v\|^2$. Every word trace for $(A, \bar{B})$ is the complex conjugate of the corresponding trace for (A, B) ; their real parts all agree. However direct multiplication gives

$$\text{tr}[A, B]^3 = 12i, \quad \text{tr}[A, \bar{B}]^3 = -12i.$$

The commutator-cube trace is a linear combination of word traces of length six. Thus the full unitary orbits differ, although both marginal spectra and all real parts of word traces agree. An actual antiunitary or CP identification, when licensed by a different comparison target, must be declared separately; it is not contained in the unitary action of Corollary 7.7.

The finite signature classifies the relative coefficient geometry at the displayed unitary resolution. A full source may additionally retain anomalous phase transport, a theta coordinate, global and boundary attachments, marked probes or a smaller lawful flavor group. Let $\mathcal{U}(Y, Y')$ be the set of unitary triples satisfying the two transformation equations in Corollary 7.7. The theorem decides exactly when this set is nonempty. Equality under the actual source comparisons additionally requires a member of $\mathcal{U}(Y, Y')$ that lifts through every retained source field. It is therefore the nonempty intersection with the actual comparison relation, not the ambient orbit alone, that decides full source identity. Degenerate stabilizers and the freedom on null complements remain part of that test; no preferred implementing unitary is selected.

7.3 Actual comparisons and structural incidence

The trace theorem separates a genuinely joint question from three simpler ones. Singular values of Y_u describe its separate left–right orbit. Singular values of Y_d do the same for Y_d . The mixed traces classify their relative position under a *common* left-unitary action. A full source comparison must also preserve the source attachments, anomaly transport, any theta data, and the admitted flavor interface. MGN6 explicitly retains rephasing, permutation and degeneracy comparisons only when they preserve the coupling witness [16, Secs. 9–12].

Proposition 7.10 (Reconstruction at a restricted comparison group). *Let a source chart have a specified lawful comparison relation on its coefficient and guard records, every member of which projects to the displayed complex-linear common-left and independent-right unitary action. After equality of the finite Gram signatures has been established, let $\mathcal{U}(Y, Y')$ be the implementing-unitary set of Corollary 7.7. The two coefficient records are equivalent in the source chart if and only if some member of this set extends to a comparison of all the remaining retained records. This criterion includes the stabilizers of repeated eigenvalues and the null-space extensions of singular maps. If every such unitary triple is source-licensed and all other fields are fixed invariantly, the finite trace signature is a complete source-orbit classifier for this coefficient branch.*

Proof. A lawful source comparison supplies the implementing unitaries together with its maps on every other retained field, proving necessity. Conversely the stipulated extension is a comparison of the full source record, not only of its matrices. It therefore proves source equivalence. The finite theorem supplies the whole nonempty implementing set, rather than selecting one unitary that might fail an additional guard. Under the last hypothesis every implementing triple has the required extension, so equality of signatures suffices. $\square$

The two reconstruction results resolve different information. Marked tensor probes recover the entire stipulated local mixed law; finitely many complex trace words recover its relative quark coefficient orbit under the stated unitary action. Neither orders the three Higgs-response roots by mass or identifies a coefficient orbit with a terminal matter certificate. Structural incidence, represented interaction and physical observation retain their distinct domains and authorities.

8 Joint role structure and compatible occupancy

A source family, its structural roles, its rooted responses and its physical occupants belong to different levels of the atlas. Their distinction is essential to the joint fermionic–bosonic structure. One charge action serves several sectors, one Higgs bridge has several incidence maps, and flavor transitions compare independently standing matter and weak-response structures. A count based on appearances in a diagram would duplicate shared sources while overlooking relational roles that have no individual particle designation.

8.1 The complete structural role portrait

The integrated Standard Model carrier theorem specifies the structural role demands, identifying the charge-response role shared by its gauge and charge sectors [8, Def. 4.1 and Lemmas 4.1–4.2]. We denote their set by $\mathcal{R}_{\text{str}}$. Its members and their locations in the present atlas are given below. A location can be a source object, an operation, or a compatibility incidence; it need not be a new carrier.

Structural role demand	Source, response or incidence carrying the role
Gauge order	The common gauge-product law, its global form and the admissible representation actions.
Transport	The source connection and its exact continuation and boundary comparison maps.
Curvature/holonomy	Curvature, loop and boundary-response operations on their admitted source domains.
Charge response	The one charge action and its shared gauge, matter and bridge restrictions.
Color transport	The color-law restriction and the distinct up- and down-colored matter action incidences.
Weak-chiral response	The weak-law restriction and the chiral matter and bridge action incidences.
Anomaly compatibility	The simultaneous representation anomaly test, with its full charge and chirality ledger.
Gauge-vacuum compatibility	The actual fixation/residual diagram and its shared gauge and Higgs compatibility maps.
Matter representation	The four kind-tagged matter diagrams and their source representation-response maps.
Color-bearing matter	The U and D kind records and their color, charge and weak incidences, with neither kind erased.
Colorless charged matter	The L kind record and its charge, weak and continuation incidences.

Structural role demand	Source, response or incidence carrying the role
Neutral weak matter	The N kind record, its weak response and the exact neutral boundary and continuation records.
Mass-response slot	The rooted incidences $\beta_{F,f}$ through C_H , before physical mass or pole coordinates are introduced.
Charge witness	The invariant charge-evaluation and continuation operations supported by the charge law.
Charge lattice	Charge normalization, global representation descent and the line-admission record when that target is requested.
Flavor transition	The relative matter/mass/weak comparison interface and its mismatch or boundary-transition record.
CP/holonomy residue	The admitted orientation, interference or holonomy residue, with its rephasing and domain conditions.
Higgs bridge/fixation	The full canonical bridge tuple, retained once, with distinct fermionic and bosonic incidence maps.

The flavor-transition and CP-residue roles carry substantive relational content even when individual matrix entries depend on presentation. The admitted mismatch compares independently standing structures across an actual common interface [7, Secs. 6–10 and App. F] [16, Defs. 3.3–3.5, Thms. 5.3, 5.5 and 13.2]. Conversely, retaining a CP-residue test does not assert a nonzero CP residue at every parameter point.

Definition 8.1 (Role portrait and positive occupancy target). The role portrait of an actual joint source is the family of the received role records in the table, their exact incident maps, and their activity, domain, compatibility and failure records. A role record retains its native source meaning; its value may include a proved absence, a partial domain or an unresolved projection status.

A positive occupancy target additionally specifies, before residue inspection, a nonempty finite family R_T of demanded standing-positive role incidences. Their required source, kind, mechanism, boundary and authority grades are fixed by the target. A status record that merely says a positive occupant is unavailable is not an eligible occupant of that positive role.

The positive role set is fixed independently of which records survive evaluation. A finer target may demand rooted incidences or terminal occupants, together with their correspondingly finer source obligations. The role portrait locates each structural demand; the occupancy problem asks whether the specified positive occupants are jointly licensed.

Proposition 8.2 (Role reception without source multiplication). *Every received primitive Standard Model role has a location in the displayed joint role portrait. The reception identifies the shared charge-response occurrence and the full shared Higgs occurrence, but does not identify distinct matter kinds, gauge mechanisms, rooted responses or terminal authority grades. Flavor and CP-residue roles are retained as relative response content. No union operation, compatibility test or gravitational constraint role introduces an additional source carrier.*

Proof. The gauge import supplies gauge order, transport, curvature/holonomy, charge, color, weak, anomaly and vacuum role graphs. The matter import supplies representation response, the color-bearing, colorless-charged and neutral-weak records, and mass-response slots. The charge import adds its invariant witness and lattice records; its charge-response graph is the same graph already received in the gauge import. The flavor import supplies the common-interface transition and surviving phase-residue records. The Higgs import supplies the one complete bridge/fixation tuple. These are precisely the role families in the source theorem [8, Secs. 3.4–4.1 and Thm. 7.1].

The earlier joint-source construction receives those graphs with their actual interface maps. In particular, its equality of the shared bridge is equality of the full intrinsic tuple and all its

incident source maps, not agreement of a label. Charge-response is likewise shared by its action and witness maps. All other identifications must preserve the relevant kind and mechanism types. A relational vertex evaluates its input sources; it supplies no new source identity constructor. The gravity role supplies the constraint and localization conditions on its received domain and enters metric reception through its separately licensed maps. Consequently it is not added to the matter, gauge or Higgs inventory. $\square$

The structural root incidence remains

$$\coprod_{f \in \{N, L, U, D\}} (I_3 \times \{r_f\}), \quad I_3 = V(K_3).$$

This is the accepted kind-positioned occurrence family, not a disjoint list of physical particles [2, Secs. 2–3]. The bosonic whole-law inventory $(1, 1, 1; 1)$ is a different grade [6, Thm. 10.6]. Neither number is a replacement for the role portrait or for a target-relative occupancy calculation.

8.2 Eligibility and the complete joint compatibility test

Fix one positive occupancy target T , after source coverage, activity reduction, quotient congruence and complete role reception. For each $r \in R_T$, let V_r be the image of its admitted standing-positive role records under the activated standing quotient. Equality in V_r is the source-certified complete target-relative equality for that role. It is not equality of a conventional designation. Let P_T be the actual complete joint-source domain and write

$$a : P_T \longrightarrow \prod_{r \in R_T} V_r, \quad D_T = \text{im } a. \quad (8)$$

Every retained relative attachment and required mixed operation appears at its incident role ports. The map therefore records a complete compatible role diagram rather than a collection of marginal response values. Source reception and reconstruction establish coverage of the actual domain; they do not place every point of the Cartesian product in D_T .

Definition 8.3 (Exact role eligibility and compatibility). Use the tagged eligible residue

$$V_T^{\text{elig}} = \prod_{r \in R_T} \{r\} \times V_r.$$

The eligibility relation is

$$E_T((s, v), r) = 1 \iff s = r.$$

The membership $v \in V_r$ already requires its complete source, standing-positive, kind, mechanism, boundary and authority typing. A role-covering pre-assignment therefore has the form $\tau_d(r) = (r, d_r)$. Its simultaneous compatibility test is

$$J_T(\tau_d) = 1 \iff d \in D_T. \quad (9)$$

Membership is witnessed by an actual admitted joint source satisfying all the retained mixed incidence, continuation, boundary and failure conditions.

The image formula specifies compatibility exactly, but does not by itself decide whether a proposed tuple has a preimage. A missing common-source witness is therefore distinct from a proof that its fibre is empty. The fibre analysis below distinguishes positive, negative and unresolved outcomes.

The tags distinguish demand-bearing records, not the underlying source objects serving them. In particular, two records can retain the same Higgs or charge source through different ports.

This is explicit target typing before role assignment, as required by the role-occupancy theorem [10, Defs. 3.11–3.16]; it is not a claim that one underlying source has become several carriers. Nor can this construction replace an independently demanded physical terminal occupant by its structural antecedent.

Theorem 8.4 (Exact eligibility and simultaneous compatibility). *On the received target, the displayed eligibility relation contains all and only the source-authorized positive role edges. The test J_T contains all and only the actual compatible complete role diagrams covered by source reception. Lawful source comparisons preserve and reflect both relations, including mixed fixed-sibling tests. In particular, separate eligibility does not imply $J_T = 1$.*

Proof. Every member of V_r is formed from an admitted role record, so its edge to r has the source’s internal or previously fixed indexical/boundary typing. Conversely, source-role coverage places every admitted occupant of that demanded grade in V_r . The tagged coproduct excludes cross-kind, cross-mechanism and cross-authority substitutions. It does not exclude an admitted shared source at a different typed port.

An actual common-source witness gives all components of $a(p)$ and all incident maps simultaneously, hence proves $J_T = 1$. Conversely, membership in its actual evaluation image supplies precisely such a source witness; coverage states that no admitted complete source diagram has been omitted. Domain-reflecting covariance transports each primitive graph, and context induction transports the entire diagram. The intrinsic fixed-sibling comparison transports the attachment map while retaining the other input itself, so a joint test cannot be replaced by a marginal one. Finally a tuple of independently eligible components need not have a common-source preimage. Only the latter is tested by (9). $\square$

8.3 Live residue, assignments and their orbits

Role-occupancy closure requires coverage of the entire live residue. The following calculation makes that requirement explicit on standing records, with presentation duplication already removed.

Definition 8.5 (Exact compatibility and role fibres). For a complete eligible tuple d and an eligible role value $v \in V_r$, put

$$F_T(d) = \{p \in P_T : a(p) = d\}, \quad F_r(v) = \{p \in P_T : a_r(p) = v\}.$$

A positive compatibility certificate exhibits a point of $F_T(d)$, with its source admission and all required role witnesses. A negative certificate proves $F_T(d) = \emptyset$ from the fixed source laws. An unresolved entry supplies neither certificate. The corresponding role-value certificates concern $F_r(v)$. They are subject to the same distinction.

Proposition 8.6 (Fibre-sensitive failure and liveness). *An eligible role value (r, v) is live if and only if $F_r(v) \neq \emptyset$. It is certified nonlive only when the source certificate proves $F_r(v) = \emptyset$. For a supplied complete tuple, $J_T(\tau_d) = 1$ if and only if $F_T(d) \neq \emptyset$. Its evaluated disposition is positive when a point of that fibre is supplied, negative when emptiness is proved, and unresolved otherwise.*

A failed necessary shared-source constraint certifies a negative result only if every point of the relevant fibre would have to satisfy that same constraint. Failure of one attempted completion does not exclude other completions in the fibre. Full role-realization completeness requires these fibre questions to be resolved throughout the claimed occupancy target; the actual-image formula alone does not discharge this requirement.

Proof. If $p \in F_r(v)$, the complete compatible tuple $a(p)$ defines a pre-assignment whose image contains (r, v) . Conversely, membership in the live union supplies a compatible pre-assignment

and hence an actual source preimage p with $a_r(p) = v$. This proves the first equivalence; the tuple equivalence follows from $D_T = \text{im } a$.

Suppose a fixed source condition $c = 1$ is necessary for every admitted source in the fibre, while the retained candidate data force $c = 0$. Any purported preimage then violates its own source admission, proving emptiness. A failed trial point proves only that this point is not a preimage unless such a universal constraint argument excludes the rest. A missing witness proves neither outcome. The three evaluated dispositions therefore record supplied proof authority, not a replacement of semantic bivalence by a third truth value. The source role-occupancy theorem expressly requires an explicit impossibility or compatibility diagnosis for every nonlive candidate [10, Def. 7.1(vi)–(vii) and Thm. 7.3]. $\square$

Theorem 8.7 (Joint live-residue computation). *The compatible pre-assignments are in bijection with D_T , and the live residue is*

$$V_T^{\text{live}} = \coprod_{r \in R_T} \{r\} \times \text{pr}_r D_T. \quad (10)$$

For rigid demanded role tags and complete standing-record equality, a compatible assignment is a bijection onto this live residue if and only if D_T is a nonempty singleton. In that case the assignment is unique, hence has one lawful assignment orbit. If $D_T = \emptyset$, there is no compatible assignment. If some $\text{pr}_r D_T$ contains two distinct standing records, no compatible pre-assignment covers the live residue.

Proof. Eligibility forces a pre-assignment to have the displayed form τ_d , and the tags make it injective. Compatibility is exactly $d \in D_T$, proving the first assertion. Taking the union of the images of all such maps gives (10).

An eligible assignment uses exactly one value in each r -fibre. It covers the union in (10) if and only if every fibre has exactly one value and the assignment is compatible. If $D_T \neq \emptyset$ and all its coordinate images are singletons, any two members agree at every coordinate, so D_T is a singleton. Its member supplies the compatible bijection. Conversely, a compatible bijection makes each fibre a singleton, giving the same conclusion. An empty image supplies no witness; two standing-distinct values in one fibre cannot both be used by the unique role assigned to that fibre. The unique assignment, when present, has one orbit under every lawful action on this complete incidence structure. $\square$

An apparently larger raw source family may nevertheless have one standing-record image. That is exactly where a source distinction is presentation or provenance rather than a further standing occupant [9, Thms. 3.4, 4.4 and Sec. 9]. Conversely, two genuinely distinct mixed records cannot be collapsed by calling the governance unique. They expose a remaining response or attachment distinction at the chosen target.

For a target with nonrigid role positions, retain the actual lawful incidence group $\Gamma_T = \text{Aut}(\mathcal{I}_T)$, where $\mathcal{I}_T$ contains the certificate, roles, live residue, eligibility, J_T , standing, tensor, boundary, continuation and failure data. Its action is

$$(\alpha, \beta) \cdot \tau = \beta \tau \alpha^{-1}.$$

The assignment space is still the space of *bijections onto the full live residue*. Quotienting a larger family of pre-assignments does not remove that requirement [10, Defs. 3.14–3.17 and Thm. 7.4]. In particular, a large symmetry of a bare graph cannot erase a standing-bearing relative flavor or boundary record.

Theorem 8.8 (Occupied fixed-datum role closure). *Let b be a complete reconstruction datum on an actual nonempty joint-source fibre. Suppose that the target-required positive role witnesses are*

present on this fibre and the reconstruction determines every retained role record and relative incidence. Then the rigid-role target has exact role-occupancy closure. If the reconstruction additionally reflects actual source comparisons, the compatible sources form one source-comparison orbit. Neither conclusion selects b among different law, flavor, global, boundary, history or state data.

Proof. Choose an actual source point over b . Its positive role witnesses give a member d^b of $D_{T,b}$, so the image is nonempty. Reconstruction fixes every native role record, its incident maps and every mixed test; hence any two points over b have the same image. More precisely, for every eligible $v \in V_r$,

$$F_{r,b}(v) = \begin{cases} P_{T,b}, & v = d_r^b, \\ \emptyset, & v \neq d_r^b. \end{cases}$$

In the first case the chosen source is a positive certificate; in the second, a purported preimage contradicts reconstruction at b . Likewise $F_{T,b}(d) = P_{T,b}$ for $d = d^b$ and is empty otherwise. Thus the fixed-datum fibre questions have exact positive or impossibility certificates, not merely an unsearched complement. Equality here is the already activated standing comparison on the retained role records; no numerical decision algorithm is asserted.

Theorem 8.7 now gives a unique compatible bijection onto the live residue. Source, quotient, role and eligibility completeness precede the calculation; reconstruction and the displayed fibre computation discharge its joint and nonlive-failure clauses. The sufficiency direction of the role-occupancy theorem applies [10, Def. 7.1 and Thms. 7.3–7.4].

For the second assertion, reconstruction equality lifts by the additional comparison-reflection hypothesis to an actual source comparison. Thus every pair lies in one source orbit. Without that hypothesis the observational conclusion still holds, but source-orbit uniqueness is not inferred. Finally the argument works in the fibre over the independently specified b ; it contains no operation selecting one such fibre from others. $\square$

Corollary 8.9 (Constructed structural role closure). *For each actual common-Higgs finite network constructed in Theorem 5.4, fix its complete source record class and demand its received positive rooted-incidence and source-law roles. That target has occupied fixed-datum role-occupancy closure, with exact exclusion of every different eligible role record at the same datum. The same conclusion holds for the positive mixed operation roles on each admitted local interaction chart of Theorem 5.6, retaining its actual coefficient and authority data.*

Proof. The first construction supplies an actual common source with the inherited positive rooted witnesses, not a tuple assembled from unrelated marginal existences. The second supplies actual nonzero mixed operation graphs on its admitted law domain. Proposition 5.10 proves complete record reconstruction and actual comparison reflection for these constructed diagrams. Their respective fixed records therefore satisfy every hypothesis of Theorem 8.8. Its fibre computation gives both the live witnesses and the certified nonlive complement. No unformed physical terminal role is included in these structural or local-operation demands. $\square$

8.4 The independently fixed positive structural-law core

The canonical single-source theorem supports a positive role target whose demands are fixed before any source point is inspected. Its occupancy is therefore independent of a subsequent choice of successful records. Occupancy of the physical projections represented in the portrait remains a separate, stronger question.

Definition 8.10 (Preformed structural-law demand). Let $R_{\text{root}} = I_3 \times \mathfrak{F}^{\text{cf}}$ be the twelve current positioned structural-role incidences of the canonical source. Let R_{law}^0 consist of the structural role

names in $\mathcal{R}_{\text{str}}$, except flavor transition and CP/holonomy residue, each at its native source-law grade. Fix the positive demand

$$R_{\text{law}} = R_{\text{root}} \amalg R_{\text{law}}^0 \amalg \{\text{boundary law, continuation law}\}.$$

The tags distinguish the positioned incidences, the global law roles, and the two interface laws. They do not count distinct underlying carriers. This demand and the following positive witness criteria are fixed before choosing s , a marking, or a complete datum.

- (i) A positioned incidence requires the actual current source restriction $\rho_f^{s,m}(i, r_f)$, retaining its kind, representation and bridge incidence, with its operator-status law. Its comparison with an accepted full FCGS record is only the explicit source-reduct comparison of (6).
- (ii) A gauge, transport, curvature/holonomy, charge, color or weak law role requires its actual source law, represented action or operation graph, with its exact boundary, domain, comparison and failure fields. Charge witness and lattice roles require the received standing charge and representation- descent ledger. No additional unreceived line-operator target is part of this demand.
- (iii) Matter and mass-response-slot law roles require the actual R_{SM} restrictions and O_Y availability, topology and status laws. An absent neutral operator is not a positive coupling: the demanded positive object is its already formed structural incidence and operator-status law.
- (iv) Anomaly and gauge-vacuum compatibility require their simultaneous source certificates on the same product, representation, charge and bridge data. The Higgs role requires the full canonical mediation tuple and its incident witness maps. A name for a common carrier or separate marginal compatibility reports is insufficient.
- (v) Boundary and continuation law roles require the actual native source interfaces and partial-operation graphs, with their composition, admission and failure laws. They do not demand a pointed history, a physical boundary state, or an unreceived metric/quantum projection.

Standing positivity here means possession of the demanded source-law or incidence witness. It does not mean a nonzero value at every argument of that law. In particular a zero anomaly obstruction, an allowed zero field value, and a partial operation domain do not remove an existing law; an unavailable law witness cannot be replaced by a status entry saying that it is unavailable.

Corollary 8.11 (Positive closure of the preformed law core). *On the canonical structural-law branch, R_{law} is simultaneously occupied by restrictions of one actual source. For each complete datum of that source at the stated law target, the fixed-datum role residue has a unique compatible full assignment and an explicitly certified nonlive complement. Thus it has exact role-occupancy closure. This neither selects a numerical law datum nor asserts a new carrier inventory.*

Proof. The demand and eligibility criteria have already been fixed in Definition 8.10. The canonical source theorem supplies $s[m]$ with the current rooted restrictions and all common source-law fields (Theorem 6.4). Its rooted restrictions supply (i), without extending the old full-target response theorem. Its actual gauge, representation and charge laws supply (ii)–(iii). Its common anomaly, vacuum-support and bridge witness ledger supplies (iv), and its received native interfaces supply (v). The structural-spine role reception in Proposition 8.2 identifies these with the preformed law role names. None was selected because a later trial happened to pass.

All these witnesses are restrictions of the same $s[m]$, so they give an actual common-source preimage, not separate witnesses for marginal eligibility. Fixing its complete law datum determines every demanded record, relative attachment and failure law. Complete-diagram reconstruction and Theorem 8.8 then compute each eligible role fibre: it is the nonempty source fibre for the

determined record and empty for every different record. The latter alternatives are excluded by the reconstruction equality, not by a missing search result. This proves full compatible coverage and failure completeness at that datum.

Whole-source reindexing transports the complete diagram and all eligibility witnesses. Fixing role positions for the calculation therefore does not select one physical family order. Nor does the argument select one datum among genuinely distinct admitted law or boundary data. $\square$

Corollary 8.12 (Witnessed flavor and CP extensions). *Fix additional positive flavor-transition or CP-residue demands before testing their population. If the same canonical source supplies every required independent carrier, common-interface, transition or residue witness, with its complete domain and relative attachments, the preformed law-core closure extends to those demands at the complete fixed datum. The mere presence of Θ_{mix} in the source tuple is not such a witness and does not establish this positive extension at every source point.*

Proof. The added witnesses give positive eligible role records on the same common-source preimage. Retaining them in the complete datum makes the same reconstruction and role-fibre argument apply. MGN6 requires independently standing comparison carriers, an actual common interface and witness-bearing mismatch or residue [16, Defs. 3.3–3.5 and Thms. 5.3, 5.5, 13.2]. A slot label or an unresolved status does not supply those fields. The complete eighteen-role portrait therefore remains valid even when this stronger positive occupancy extension is unavailable. $\square$

A neutral terminal child, a charged-lepton pole target, a colored terminal target and an unrestricted physical common apex require different positive witnesses. The structural-law result leaves unformed terminal branches unformed: a status entry is not a positive occupant of the corresponding physical role [2, Secs. 4, 6 and 8][6, Secs. 7–8 and 11].

Corollary 8.13 (Cardinality is graded and downstream). *Where positive role-occupancy closure is established, the live standing-record cardinality equals the independently fixed role cardinality. The structural demand portrait, kind-positioned rooted incidences, bosonic whole-law inventory, response-support dimensions and physical projection multiplicities are not interchangeable cardinalities.*

Proof. The first equality follows from the compatible bijection, not from the presentation of a list. The other quantities count different source, incidence, response or projection objects; their forgetful maps are not identifications. In particular the same C_H is used at several demand ports, and one rooted carrier can have several physical modes. Neither reuse nor projection changes its source identity. $\square$

9 Physical projection, conventional labels and structural surplus

Conventional Standard Model designations appear at the physical projection level of the structural atlas. Their interpretation depends on source ancestry, projection authority and degeneracy conventions. Making those dependencies explicit preserves the usefulness of particle names without treating nomenclature as a source of identity.

9.1 Projection domains, reconstruction and faithfulness

Let A_T be the relevant activated structural response quotient. An authority packet α retains the physical target, metric licence, source and continuation witnesses, allowed degeneracy, boundary and calibration data, comparison maps and provenance. Write

$$p_\alpha : D_\alpha \longrightarrow P_\alpha, \quad D_\alpha \subseteq A_T,$$

for its admitted physical projection. Several physical modes over one structural response are described by the full projection correspondence or by a mode-indexed domain; no unique mode is selected implicitly. The following results apply on each exact map domain. The complete descent record retains the upstream argument and α , as in the established carrier-preserving projection [2, Defs. 6.1–6.2 and Thm. 6.3].

Definition 9.1 (Projection authority grades). A projection entry states separately:

- (i) whether its complete source and authority packet is supplied;
- (ii) its exact domain and any proved empty or partial subdomain;
- (iii) whether its output is a structural representation, a licensed classical realization, or a supplied physical channel;
- (iv) whether its target comparisons are faithful; and
- (v) which requested target observations its output reconstructs.

An absent authority packet has no inferred projection value. A proved empty domain and an unformed domain are different entries.

Theorem 9.2 (Exact projection reconstruction criterion). *Let $\mathcal{O}$ be the fixed family of requested observations on D_α , including their domains and failure values. Every $o \in \mathcal{O}$ factors through p_α if and only if*

$$\ker p_\alpha \subseteq \bigcap_{o \in \mathcal{O}} \ker o. \quad (11)$$

The factor is unique on the actual image of p_α . The projection is injective on the structural quotient exactly when $\ker p_\alpha$ is equality. Observational completeness for a smaller target does not in general imply that injectivity.

Proof. A factorization $o = \bar{o}p_\alpha$ makes equal projected values have equal observed values, proving necessity. Conversely (11) makes $\bar{o}(p_\alpha(x)) = o(x)$ independent of the preimage. Every element of the actual image has a preimage, so the factor exists there and is unique. Encoding definedness and failure as part of the observed value proves the same statement for partial tests without silently enlarging their domains. Injectivity is the defining equality-kernel criterion. A smaller observation family may deliberately fail to separate two structural classes, so completeness for that family need not imply injectivity. $\square$

For groupoid-valued presentations, faithfulness additionally concerns arrows. Equality of object-level invariants does not prove that every presentation comparison lifts to an actual source arrow; that is the separately required comparison reflection used in the joint reconstruction theorem. The retained ancestry makes the *full descent record* carrier-preserving even when its physical coordinate or its conventional name alone is not faithful.

9.2 A typed conventional dictionary

For fermion kind f , the dictionary has the source order

$$\text{Disc}(E_f) \xrightarrow{\eta_f} \mathcal{S}_f \xrightarrow{\chi_f} Q_f^Y \xrightarrow{\beta_{F,f}} \mathcal{H}_{F,f}(C_H) \dashrightarrow P_{f,\alpha} \xrightarrow{\ell_{f,\alpha}} \text{Name}_{f,\alpha}. \quad (12)$$

The partial factor is defined only with its complete physical authority. For gauge and Higgs labels, the corresponding ancestry runs through their source-law, fixation/residual and mechanism records before reaching a physical channel and its naming map. These constructions preserve the different bosonic projection mechanisms [6, Secs. 2, 5 and 8].

Conventional designation	Structural ancestry	Authority-sensitive distinction
ν_e, ν_μ, ν_τ	Neutral weak matter and its rooted response/boundary record	Weak-flavor endpoint labels are not automatically transported propagation eigenstate identities.
e, μ, τ	Colorless charged matter, rooted Higgs response and charged physical projection	A mass order or pole designation does not form an upstream root or terminal certificate.
u, c, t	Up-colored matter with its color, charge, weak and rooted Higgs incidences	Its full target is not the down-colored target even when their root reducts are isomorphic.
d, s, b	Down-colored matter with its distinct typed incidences	A common three-position display does not identify its charge or confinement record with the up-colored record.
γ	Charge/weak gauge ancestry and the electromagnetic residual interface	A supplied polarization or massless channel is later than the premetric charge law.
$W^\pm, Z$	Weak/charge/Higgs fixation and its response decomposition	Massive response support and an unstable physical pole require their respective source authorities.
g	Color gauge transport and curvature response	Generator components, a color field and a confined projection are not independent asymptotic particle carriers.
h_{phys}	The one bridge/fixation carrier and an admitted scalar response	A physical resonance is not the source definition of C_H .
Predicted HNL	The independently certified neutral heavy support on its exact child scope	The rank-three heavy sector is not an additional generation root.

The dictionary records ancestry while preserving the exact physical readiness of each branch. The neutral terminal children remain distinct from their parent complement, and familiar charged or colored names do not strengthen their terminal dispositions [2, Secs. 4 and 6]. A color field does not by itself define a color-charged Wigner particle; a pole criterion does not ensure a nonzero residue in every permitted channel.

9.3 Generation markings are not generation identities

The unmarked occurrence object I_3 carries the external permutation action of S_3 . Let

$$M = \text{Bij}(I_3, \{1, 2, 3\})$$

be its marking torsor. A marking is a display coordinate, not an extra occurrence. Suppose $n_f(1), n_f(2), n_f(3)$ are the three designation slots of an independently licensed dictionary for a given kind and projection target.

Theorem 9.3 (Equivariant naming and absence of a canonical family order). *The naming rule*

$$L_f(m, i) = n_f(m(i))$$

is invariant under simultaneous change

$$(m, i) \mapsto (m\sigma^{-1}, \sigma i), \quad \sigma \in S_3.$$

There is no injective S_3 -invariant map from unmarked I_3 to a three-name set with trivial action. Consequently a three-family display requires an explicit marking or an independently standing ordering witness. It does not quotient I_3 to one root and does not supply a physical mass order.

Proof. The composite marking evaluates to $(m\sigma^{-1})(\sigma i) = m(i)$, proving the first claim. If $l : I_3 \rightarrow N$ were invariant and $i, j \in I_3$, transitivity supplies σ with $\sigma i = j$. Then $l(j) = l(\sigma i) = l(i)$, so l is constant and cannot be injective. Retaining the pair (m, i) is different from identifying all roots: its simultaneous-action orbits are distinguished by $m(i)$. An additional ordering witness can restrict lawful comparisons, but that witness must be received at its own target and cannot be inferred from the mere existence of the torsor or from a downstream fitted mass table. $\square$

The theorem concerns the unmarked structural reduct. On a full source, fixed boundary, coupling or relative-flavor data may restrict the action to a smaller preserving subgroup. A simultaneous re-presentation transports attachments together with labels. By contrast, permuting one attached source while holding its attachment fixed can change a genuine mixed response. That change is relative flavor content, not an additional occurrence root [7, Sec. 8.3][16, Defs. 9.1–9.3 and Thms. 9.5–9.6].

Theorem 9.4 (Conventional-label factorization). *Every conventional label admitted at the fixed target factors through its registered source/role record, any applicable rooted response, its complete projection authority and its lawful marking. The full labelled descent record preserves the upstream kind and carrier ancestry. Bare labels are neither required to separate all structural responses nor permitted to create missing source or projection authority.*

Proof. For fermions the received source construction is the literal chain (12); its first maps are the accepted rooted chain, and its only physical factor has the stated domain. The lawful marking changes according to Theorem 9.3. For bosons the source, fixation and mechanism records determine which physical projection is being requested, and the naming map is the last map of that projection's record. A finite mixed process uses the labels of its actual typed endpoints and its supplied process projection, not a new label-first source constructor. Source and use coverage exhaust these admitted records.

Forgetting the final designation and later physical coordinates recovers the recorded upstream argument; thus its kind and ancestry are preserved. A naming map can identify several presentation modes or response records, so no injectivity is inferred from nomenclature. Outside a partial factor's domain there is no composite value. Neither a familiar name nor its place in a printed table supplies that missing factor. $\square$

9.4 Exhaustion of primitive label surplus

Let X be the fixed actual source domain and $\Sigma : X \rightarrow Q = \text{im } \Sigma$ its complete mixed-use signature. For a proposed formed partial labelling map $K : D_K \rightarrow N$, $D_K \subseteq X$, retain its definedness and its declared failure diagnostic in the totalized value $\widehat{K} : X \rightarrow \widehat{N}$. In particular defined and undefined values have disjoint tags. This totalization encodes the semantics of the already formed domain. It supplies neither a missing authority packet nor an effective decision procedure for every domain question.

Proposition 9.5 (Global label descent and its finest coarsening). *The complete proposed label record descends to the fixed atlas if and only if*

$$\ker \Sigma \subseteq \ker \widehat{K}. \quad (13)$$

When it descends, its factor $\overline{K} : Q \rightarrow \widehat{N}$ is unique. Thus both the labelling domain and every retained failure distinction must be constant on each complete-signature fibre, not merely on some witness-separated pairs.

More generally, let $L = \text{im } \widehat{K}$, and let $\approx_K$ be the equivalence relation on L generated by

$$\widehat{K}(x) \approx_K \widehat{K}(y) \quad \text{whenever } \Sigma(x) = \Sigma(y).$$

The quotient $q_K : L \rightarrow L/\approx_K$ is the finest quotient of attained label values for which $q_K\widehat{K}$ descends to Q : every postprocessing $h : L \rightarrow Z$ with $h\widehat{K}$ factoring through Σ factors uniquely through q_K . This coarsening diagnoses the redundant label distinctions; it does not certify the original K or restore an erased domain or failure distinction.

Proof. If $\widehat{K} = \overline{K}\Sigma$, equal signatures give equal totalized label values. Conversely (13) makes $\overline{K}(\Sigma(x)) = \widehat{K}(x)$ independent of the representative. Every point of Q has a representative, which proves uniqueness. Disjoint domain tags and the retained failure values are part of $\widehat{K}$, so the same argument applies to them.

The quotient identifies every generating pair, making $q_K\widehat{K}$ constant on each signature fibre and hence factorable by the first argument. If $h\widehat{K}$ factors, h identifies each generating pair. Equality is transitive, so h identifies their generated equivalence relation and has a unique factor through the surjection q_K . A quotient that identifies a success tag with a failure tag has explicitly forgotten that distinction; it cannot be used as an exact replacement for the original partial labelling claim. $\square$

Theorem 9.6 (Same-scope label-surplus exhaustion). *After complete source reception, the mixed-use quotient and role portrait have been established, a formed proposed label record has the following exhaustive dispositions:*

- (i) *If (13) holds, the entire record factors uniquely through the registered atlas and adds no independent same-scope identity classifier.*
- (ii) *If that inclusion fails while the target and its licensed uses remain fixed, the distinctions within complete signature fibres may be retained only as presentation or provenance skin. The finest descended coarsening is exactly $q_K\widehat{K}$; the uncoarsened record need not itself factor, even if it also records some genuine atlas distinctions.*
- (iii) *it requires a new source, witness, role, continuation, boundary or authority domain and belongs to a target extension;*
- (iv) *If it asserts standing authority for a distinction inside an unchanged complete-signature fibre without such a target extension, it fails as a same-scope classifier promotion.*

There is therefore no additional primitive same-scope carrier whose only distinguishing content is a conventional label. This conclusion preserves genuine witnessed relative flavor, CP-residue, global or boundary distinctions; it does not declare them absent because their individual coordinates are nonprimitive.

Proof. First separate proposals that change the source or target interface; these give (iii) and require fresh activation at that interface. For a formed proposal on the fixed domain, test (13) on every actual signature fibre. If it holds, Proposition 9.5 constructs the global factor, proving (i).

If it fails, there exist $x, y \in X$ with equal complete signatures and different totalized label records. Every already licensed observation and continuation agrees on that pair by mixed-use completeness. More generally, this is true of every generating pair of $\approx_K$. Keeping those label distinctions without new standing authority is therefore skin relative to the fixed target. The universal property of q_K gives the precise global descended content, proving (ii); it does not falsely infer factorization of the original record from a distinction visible on some other pair.

If one of these within-fibre distinctions is instead asserted to have standing under the unchanged interface, it contradicts the activated uniqueness of the standing quotient and introduces an independent same-scope classifier, proving (iv) [9, Thms. 3.4, 4.4 and App. C]. Making that distinction observable by adding a new admitted use changes the interface and returns to (iii), not to a repair under the old target. These cases exhaust the global kernel inclusion and the only two possible uses of its within-fibre surplus. They agree with the source structural classification

of label and witness surplus [11, Thms. 11.1 and 18.1, Prop. 14.1]. Every original atlas distinction remains in Σ throughout. Thus taking the descended quotient of a label does not erase genuine mixed, boundary or CP-residue content from the atlas itself. $\square$

Corollary 9.7 (Heavy neutral support does not create a fourth root). *On the received ready neutral child scopes, the parallel rank-three heavy sector and the three light sectors retain their certified neutral carrier ancestry. Naming the heavy projection a predicted HNL neither creates a fourth member of I_3 nor duplicates the one C_H . Numerical search annotations remain annotations at their own authority grade.*

Proof. The accepted neutral result separates occurrence roots, rooted response slots, one bearer and its light/heavy physical decomposition [2, Secs. 4 and 6]. The dictionary preserves those fields. The final designation cannot alter the domain of an earlier root constructor, and the heavy degeneracy orbit is not a collection of new roots. The same argument applies to a search band or response-grid annotation: it reports information downstream of the neutral type theorem and does not supply additional identity authority. $\square$

10 Typed processes and the gravitational interface

Interactions connect endpoints without identifying their carriers. A weak transition between up-type and down-type responses is not a full-target isomorphism of their source kinds. Production, subsequent decay and detector recording likewise remain distinct typed operations, even when they belong to one experimental event.

10.1 Process composition and physical authority

Let the endpoint alphabet contain the admitted matter and bosonic interface sorts, their duals and orientations, as well as independently received composite, boundary and record sorts. A process profile specifies a target, input and output words, all relative source attachments, and an ordered word in the received process types. These retain transition, weak action, pair production, propagation, mode conversion, self-interaction, decay, splitting, confinement continuation, hadronization, restoration and record formation. Coincident names in the fermionic and bosonic process families are identified only through an actual same-type source comparison.

The witness over a profile contains its actual operation graphs, continuation, boundary, authority and first-failure records. These are the fermionic process fibres [2, Sec. 7] and bosonic source diagrams [6, Sec. 9], together with the mixed vertices of Section 5. A law-level graph, a state or scattering channel, and a measurement reconstruction have distinct authority levels. State, scattering and measurement fibres are populated only by their actual source maps.

Theorem 10.1 (Joint exact-domain process atlas). *The admitted joint process diagrams form a typed partial composition system with identities. Composition is associative in its value, exact domain, ordered process-type word and first-failure record. Full-profile source comparisons act functorially; the process-witness categories over the profile groupoid form a split fibration when transported records are taken as the chosen pullbacks. Evaluation descends through the complete mixed observational congruence on its exact domain. The populated fibres are precisely those supplied by the received source witnesses.*

Proof. For $f : D_f \subseteq X \rightarrow Y$ and $g : D_g \subseteq Y \rightarrow Z$, the composite domain is

$$D_{gf} = \{x \in D_f : f(x) \in D_g\}.$$

For a third map h , both bracketings require $x \in D_f$, $f(x) \in D_g$, and $g(f(x)) \in D_h$, and have the same final value. The first failed vertex is the same in the retained evaluation order.

Grafting higher-arity diagrams uses the same test at each matching actual port, including its common-source predicate. Concatenation preserves the full type word and intermediate boundaries.

Transport of a witnessed profile pulls back every vertex and incident map. Primitive domain reflection and fixed-sibling covariance show that this transport commutes with composition. Transported records compose strictly, giving chosen identity and composite pullbacks. The Grothendieck construction of these pullback functors has the canonical cartesian lifts, which proves the split-fibration statement. If charts are instead retained only up to isomorphism, the same construction gives the equivalent, not necessarily strictly chosen, fibration.

Every evaluation is a received context. Theorem 4.1 therefore gives its quotient descent, including the domain and failure record. No step creates a state, pole or boundary operation absent from the source, so the population assertion is exactly the actual source image. $\square$

Contraction has a corresponding elementary constraint. For an invariant scalar vertex on central-character and unbroken Abelian-weight components, a nonzero value requires trivial product central character and zero total Abelian weight. Acting by the relevant central element or circle multiplies the output by that character, proving the requirement. Contraction with an admitted dual cancels the two internal characters, so the requirement survives on external ports [6, Thm. 9.3]. These necessary conditions do not prove a nonzero amplitude. The non-Abelian, spin, coefficient and continuation conditions remain.

Population is determined at the specified authority level. The polynomial chart has nonzero gauge-matter and charged Yukawa vertices, while the ready neutral children have their target-specific process witnesses. The U/D terminal targets do not acquire a confinement-complete packet from a color-response vertex, and an unstable channel does not acquire a proper lifetime from a diagrammatic decay label. Stronger source maps retain their actual domains; the combined process grammar neither removes those maps nor enlarges their domains.

10.2 Premetric gravity as a condition of physical realization

Physical realization also requires the structural conditions under which an admissible interior supports reusable observations and continuation. In the regular Hamiltonian comparison class, take a symplectic carrier (S, ω) , a regular standing surface C , and a quotient submersion $\pi : C \rightarrow S_{\text{phys}}$. Physical observables are pullbacks along π . The exact AER conditions require extension-independent Poisson evaluation, quotient separation and cotangent spanning, and no nonzero presymplectic tangent class invisible to every physical observable [18, Def. 5.12].

Proposition 10.2 (Constraint necessity at the realization interface). *At that regular AER interface,*

$$(TC)^{\perp\omega} \subseteq TC, \quad \ker(d\pi) = \ker(\omega|_C).$$

Consequently standing is coisotropic and source redescription is precisely its characteristic distribution. Regular local defining constraints have first-class closure. With the source's presentation-arbitrariness condition, the Hamiltonian has the reduced-plus-multiplier form. These structures are necessary for this physical interior; they are not extra matter or bosonic carrier positions.

Proof. Let $\xi \in (T_x C)^{\perp\omega}$. Choose a local function H vanishing on C with $dH_x = \omega_x(\xi, \cdot)$. Its Hamiltonian vector at x is ξ . Extension independence applied to two functions vanishing on C makes their bracket vanish there. Since the differentials of such functions span $\text{Ann}(T_x C)$, ξ is tangent to C . Applying the same condition to a physical observable shows that every physical observable differential annihilates ξ . Cotangent spanning then places ξ in $D_x = \ker d\pi_x$.

Conversely, if $v \in D_x$ were not in $\ker(\omega|_C)_x$, maximal same-scope observability would give a physical O with $dO_x(v) \neq 0$. But $O = O\pi$ implies $dO_x(v) = 0$. Thus $D = \ker(\omega|_C)$, using

the established coisotropic inclusion. Regular coisotropic normal form supplies local defining constraints ϕ_a . Their Hamiltonian fields are tangent to C ; hence $\{\phi_a, \phi_b\}$ vanishes on C . The regular defining-function lemma gives $\{\phi_a, \phi_b\} = \sum_c C_{ab}^c \phi_c$. For a presentation Hamiltonian, its restriction descends to the reduced Hamiltonian h . The difference from an extension of π^*h vanishes on C , hence equals $\sum_a u^a \phi_a$. These are the exact source steps [18, Thms. 6.3, 7.5, 7.8 and 8.2]. They concern standing, quotient and continuation, not a carrier-formation operation. $\square$

The necessity has a direct operational meaning. Without this constraint role, a physical observable may depend on its extension off the standing surface, physically relevant directions may be invisible to the quotient, or continuation may fail to preserve standing. Any such failure compromises the claimed physical interior itself. This structural result leaves the numerical gravitational coupling and the choice among metric dynamics to their respective realization questions; it does not identify every constrained system with one metric theory.

10.3 Mixed metric response and exchange

Inside a licensed smooth metric envelope let the actual source have a finite decomposition of its full continuous linear compact metric first response,

$$W_{\text{tot}}(k) = \sum_a W_a(k), \quad W_a(k) = \frac{1}{2} \langle \mathsf{T}_a, k \rangle.$$

Every interaction, vacuum and boundary contribution required by the target is counted exactly once. The fermionic and bosonic terms share the metric and its allowed variations. Compact interior vector-field tests isolate the local exchange identity; physical boundary flux remains a separate response.

Theorem 10.3 (Joint matter–boson exchange closure). *Suppose the actual source exchange identities are*

$$W_a(\mathcal{L}_X g) = - \left\langle J_a^{\text{ext}} + \sum_e B_{ae} j_e, X \right\rangle, \quad \sum_a B_{ae} = 0.$$

Then its unique total metric response satisfies

$$\mathsf{T}_{\text{tot}} = \sum_a \mathsf{T}_a, \quad \nabla^\mu T_{\mu\nu}^{\text{tot}} = \sum_a J_{a,\nu}^{\text{ext}}.$$

It is conserved exactly when the total external exchange vanishes. Source comparison preserving the metric tests and mixed incidences preserves this identity. Sectorwise conservation is not required.

Proof. Finite sums preserve distributional representation. For a symmetric response, duality and integration by parts give $\frac{1}{2} \langle \mathsf{T}, \mathcal{L}_X g \rangle = - \langle \text{div } \mathsf{T}, X \rangle$. Summing the supplied exchange identities cancels each internal current by the column-balance condition. Equality on every compact test proves the displayed distributional identity and both directions of the conservation criterion. Actual source comparisons intertwine each response, test and incidence, so intertwine their sum. This is the received coupled Ward argument applied to the new matter–boson source [6, Thm. 9.5]. $\square$

A covariant joint action gives the exchange identity by first variation. Its Euler–Lagrange remainders vanish on a joint solution and remain explicit in an off-shell polynomial chart. A potential minimum or a sum of source terms is therefore distinct from a solution of all coupled equations. For an admitted physical realization, the gravitational interface receives one total response, including internal exchange, rather than independently assigned sources for individual particle designations.

11 Complete reconstruction and no-repair minimality

A minimal structural description retains precisely the information needed to distinguish its target observations. Importance and irredundancy are different: a shared Higgs record can be essential to the theory yet redundant in a description that records it twice. Conversely, a short particle list may omit an observable relative phase, source boundary or failed operation.

11.1 Reconstruction across the admitted levels

For a complete joint source let P_T denote the domain of its reconstruction records. Such a record retains the full source-family and relative-chart data; root, kind and role incidences; one canonical bridge and its distinct response maps; actual mixed-operation tensors; and all independently required global, continuation, boundary, authority and failure data. On the polynomial branch, Section 7 represents the stipulated law by coefficients that reconstruct it. Source fields without an established reduction retain their original form. The record is therefore not restricted to a finite list of numerical parameters.

An admitted represented-state target additionally requires its complete joint kernel. Fermion-boson words retain graded operator order and the actual source relations; the two marginal state kernels alone do not determine this mixed information.

Proposition 11.1 (Received graded-state reconstruction). *Suppose two actual represented joint sources have word vectors Ψ_w, Ψ'_w with dense spans, identical complete mixed Gram kernels and identical admitted word relations and actions. Suppose also that each compared bounded boundary map intertwines on the dense word span under its retained target comparison. The assignment $\Psi_w \mapsto \Psi'_w$ extends to a unique unitary. It intertwines the specified word-span actions, the closed operators for which these spans are actual common cores, and separately supplied complete operator graphs when their comparison is retained. It preserves the graded relations and bounded boundary maps. Equality of marginal kernels alone does not imply these conclusions.*

Proof. The squared norm of a finite word combination is $\sum_{w,v} \bar{c}_w c_v K(w,v)$. Equal kernels identify exactly the null combinations and give an isometry of the two spans. Density gives the unitary extension. Equal word actions give intertwining on that span. An actual common core determines the closed graph by closure; otherwise the full graph is independently retained. Bounded boundary maps extend by continuity. Graded signs are among the word relations, so are preserved by the same map rather than replaced by commuting relations. This is the complete joint-kernel argument [6, Thm. 10.1], with the newly received graded words.

For the last assertion, even two finite two-level systems admit the pure vector $(|00\rangle + |11\rangle)/\sqrt{2}$ and the mixed state $I_4/4$ with identical marginal states. Their product- σ_z expectations are respectively one and zero. The example diagnoses the informational implication; it does not assert that either finite model realizes the Standard Model state target. $\square$

Theorem 11.2 (Integrated mixed reconstruction). *Every admitted structural, response, continuation, projection and process use of the received target factors through its complete reconstruction record, with exact domain and failure value. At a fixed record there is no further independent choice of an admitted response. The full observational quotient is the actual evaluation image of these records. Source-orbit reconstruction additionally has the comparison-reflection condition of Proposition 4.4.*

Proof. For a structural vertex use its complete received source graph and incident maps. For a polynomial law vertex use Theorem 7.3. For a represented state vertex use Proposition 11.1 on precisely its core or full-graph domain. Upstream continuation retains the independently received source category and its complete requested tests. The remaining source fields retain the

physical-authority and boundary conditions of a stronger projection, whenever present. Thus each primitive value, domain and diagnostic is determined. Source-use coverage and finite context induction determine every received use. Theorem 4.1 identifies the resulting quotient with that evaluation image. Object reconstruction without lifted comparisons does not establish source-orbit identity, which accounts for the final clause. $\square$

11.2 Extensional completeness beyond the expression calculus

Contextual completeness concerns the received operation calculus. Complete source reconstruction yields a distinct extensional result: an independently admitted use need not be expressed in that calculus to factor through the reconstructed source class.

Theorem 11.3 (Factorization of every source-covariant use). *On a source branch whose complete record reconstruction reflects and lifts actual source comparisons, let*

$$R_{\text{src}} : P_T \longrightarrow B_{\text{src}}$$

be its attained complete-record isomorphism class. Every independently admitted source-covariant partial use u , including its exact domain and intrinsic failure value, factors uniquely through R_{src} . The use need not be an expression in the earlier finite calculus.

If the complete observational signature also reconstructs R_{src} on this branch, every such use factors through that signature and cannot refine its quotient. Without this additional source-resolution property the universal conclusion is about R_{src} , not automatically about a coarser observational quotient.

Proof. Suppose $R_{\text{src}}(p) = R_{\text{src}}(p')$. Complete record reconstruction and comparison lifting supply an actual source isomorphism between p and p' . Domain-reflecting covariance makes u defined on one exactly when it is defined on the other, and identifies its intrinsic output or retained failure. Thus the totalized use is constant on every R_{src} -fibre. The actual-image factorization argument constructs the unique factor. This proof uses the independently admitted use and its source covariance, not a declaration that it belongs to a syntax generated by selected operations.

For the second assertion compose with the reconstruction of R_{src} from the signature. Conversely, if the signature loses a source distinction, source covariance alone does not make that distinction invisible to every possible admitted use. Therefore the stated extra hypothesis cannot be discarded. $\square$

Complete diagram records satisfy the source-reconstruction premise by Proposition 5.10. The canonical source likewise retains all indexed tensors, incidence maps and lawful comparisons. On the marked finite-law branch, coefficient probes recover the variable operation tensors. Full source resolution also requires the independent native fields; a smaller experimental readout need not have that resolution.

The two completeness claims thus have different scopes. The complete contextual signature is final for its declared operational target; the complete source-class record is final for every independently admitted covariant use of the same source. They coincide when the explicit source-resolution test holds. Neither conclusion supplies a new physical use or a missing authority packet.

11.3 An exact deletion test

Write $R : P_T \rightarrow \mathcal{V}_T$ for the totalized complete response signature. A proposed compression is an actual map $d : P_T \rightarrow D$, evaluated at the same source record, not at a newly selected point.

Theorem 11.4 (No-repair compression and deletion). *The compression d preserves all observations if and only if*

$$\ker d \subseteq \ker R.$$

When this holds, the unique map on $\text{im } d$ satisfying $R = \bar{R}d$ reconstructs the complete response. A failed inclusion is equivalent to two actual records with equal compressed data and different full observations. Such a pair cannot be repaired by changing a sibling input, source point, target domain or comparison relation while still claiming losslessness of this compression.

Proof. A factorization makes equal d -values have equal R -values. Conversely the kernel inclusion makes $\bar{R}(d(p)) = R(p)$ independent of the actual representative; surjectivity onto $\text{im } d$ gives uniqueness. Its failure is exactly the stated separating pair. Choosing another record does not give both originally different values to the same d -value. It changes the problem instead of constructing the required function. $\square$

Proposition 11.5 (Finite deletion-minimal packets). *Suppose a finite indexed family of fields $(p_i)_{i \in I}$ reconstructs R on the actual source domain. There is an inclusion-minimal subfamily that still reconstructs R . Each retained field of such a subfamily has an actual separating pair when it alone is deleted. A source family can have several distinct minimal subfamilies; inclusion-minimality does not imply minimum cardinality or uniqueness of a presentation.*

Proof. Among the subsets of the finite set I whose coordinate maps satisfy the kernel inclusion, choose an inclusion-minimal one. At least I itself qualifies. Deleting any retained index breaks the inclusion, so Theorem 11.4 supplies its separating pair. Different subsets can encode the same information, so neither of the additional uniqueness claims follows. $\square$

This is an existence result for finite field families, not an algorithm for deciding arbitrary source equalities. An infinite family need not contain a minimal reconstructing subfamily. The owned-interface minimality of the fermionic source [2, Sec. 9] also differs from observational deletion-minimality. A theorem may make an indispensable contribution to the source interface even when its resulting information is represented elsewhere as well.

11.4 Minimal coefficient data for the mixed law

Proposition 11.6 (Charged coefficient irredundancy). *On the marked polynomial-law branch with freely variable charged matrices and all other laws fixed, the complex entries of Y_u, Y_d, Y_e are a reconstructing, deletion-minimal coefficient packet. Equivalently, their real and imaginary parts form a deletion-minimal real packet. For three marked copies this is twenty-seven complex, or fifty-four real, coefficient fields. This is a law-coordinate count, not a count of physical parameters after source comparison, and not a count of carriers.*

Proof. Lemma 7.2 proves reconstruction. Change one complex entry while holding all other entries fixed. Both are actual multilinear operations in the specified polynomial-law domain. The corresponding ordered monomial coefficient changes and its elementary probe separates them, while every undeleted coordinate agrees. The same construction varies one real or imaginary part. There are three matrices with nine complex entries each. No standing-source population beyond that polynomial domain is used for the count. $\square$

Relations on a source-restricted standing image can make coefficients dependent. The same packet reconstructs the law, but minimality must be determined on that actual image rather than transferred from the freely variable chart.

Under the full common-left unitary comparison, the finite trace-word packet of Theorem 7.6 reconstructs the two Gram matrices up to simultaneous equivalence. The list need not

be irredundant. The positive Gram examples in Section 7 nevertheless establish two precise information losses: separate spectra omit relative flavor information, while the real parts of all trace words can omit a complex mixed invariant. Both examples lie in the stated coefficient-law domain. Their application to a more restrictive physical source requires that source to admit both comparison points.

11.5 Shared sources and distinct incidences

The integrated description retains one complete Higgs source record, one shared charge-law record and the actual incidence maps from each demand port. The fermionic and electroweak Higgs-response maps have different operand and output types and are not duplicates. Repeating the identical Higgs record in each sector, by contrast, adds no observation. Eliminating that repetition preserves the distinct incidences.

Gravity constraints, joint compatibility, response-support dimensions, contraction multiplicities and experimental search annotations have their own types; none adds another carrier. An annotation is necessary only for a reporting target that observes it. An exact boundary or failure field is necessary whenever its deletion has the separating pair of Theorem 11.4. The target-relative criterion preserves the required scientific content without imposing a unique presentation size.

12 The structural Standard Model atlas

The results determine an atlas of sources, incidences, responses and projections with specified authority. The familiar particle display is a downstream presentation of this object. The atlas also retains information that the display leaves implicit: common charge and Higgs support, relative flavor, exact mixed-operation domains, confinement boundaries and the constraint conditions of physical realization.

12.1 The assembled object

Let $\mathcal{B}_T$ be the actual image of the independent source and boundary data of the received target. Over each $b \in \mathcal{B}_T$, form

$$\mathfrak{A}_T(b) = \left(\mathcal{C}_{\text{SM}}^{\text{str}}, \mathcal{S}_{\text{SM}} \sqcup \mathcal{R}_{\text{SM}}, Q_T, \mathcal{I}_T, \mathcal{E}_{\text{Proc}}, \Pi_{\text{typed}}^{\text{phys}}, \mathcal{F}_T \right)_b.$$

Here the first entry is the graded carrier/source architecture; the second retains the different fermionic and bosonic sector/response mechanisms; Q_T is the complete mixed observational quotient; $\mathcal{I}_T$ is the role-incidence and compatibility structure; $\mathcal{E}_{\text{Proc}}$ is the typed process atlas; $\Pi_{\text{typed}}^{\text{phys}}$ is the family of actual authority-indexed partial projections; and $\mathcal{F}_T$ retains their source-defined domains, absences, failures and unresolved authority records. This notation does not promote the entries to independent carriers.

The branch base is an actual joint source image, not the product of independently populated marginal branch lists. A common diagram can retain several lawful local charts, but it retains their actual overlap maps and every shared restriction. A classical history or quantum state is an additional source attachment where supplied, not a value selected to populate a formal product.

Theorem 12.1 (Structural carrier–response classification). *For the received structural Standard Model source target and its admitted successor operation domains, the atlas $\mathfrak{A}_T$ has the following properties.*

(i) *The accepted fermionic root and response chain is retained literally at each kind:*

$$E_f = I_3 \times \{r_f\}, \quad \text{Disc}(E_f) \xrightarrow{\eta_f} \mathcal{S}_f \xrightarrow{\chi_f} Q_f^Y, \quad |I_3| = 3.$$

Full-target kinds remain distinct, and the external root permutation is not an orbit quotient of the occurrence set.

- (ii) *There is one canonical Higgs source with distinct fermionic and electroweak incidence maps. The canonical structural common-source construction evaluates the demanded shared matter, gauge and vacuum laws together. The represented polynomial branch also contains actual nonzero mixed gauge and charged Yukawa operations.*
- (iii) *Complete source reception, fixed-sibling covariance, continuation and failure tests precede the mixed quotient. That quotient is the greatest domain-reflecting congruence for the received target. Its exact refinement of the separate-atlas tuple is $\ker \Sigma_T = \ker m \cap \ker \mu$.*
- (iv) *Complete reconstruction determines every received observation on its actual domain. Marked coefficients reconstruct the stipulated finite local law; complex trace words of length at most seventeen reconstruct the simultaneous three-dimensional quark Gram orbit under the specified unitary comparison. Stronger source comparisons retain their additional lifting conditions.*
- (v) *The eighteen received structural role demands have the explicit portrait of Section 8. Eligibility, simultaneous compatibility, live residue and assignment orbits are calculated at their demanded grades. The independently preformed structural-law core is simultaneously occupied and has exact closure at each complete source datum. Additional positive flavor, CP-response and physical terminal targets keep their separate same-source witness requirements.*
- (vi) *The process and projection maps preserve source ancestry, authority, exact domains and failure records. Conventional labels factor through their typed descent records and lawful markings. Additional primitive same-scope label authority is exhausted by the global label-kernel theorem.*
- (vii) *A compression preserves the full target exactly when its kernel lies inside the full observational kernel. The stated finite minimal-packet and coefficient irredundancy results apply on their exact source domains; duplicate storage and distinct scientific incidences are treated differently.*
- (viii) *No projection, process, fit, support rank or compatibility predicate alters the established source identity or fills a missing upstream witness. Every inherited partial or split branch remains present with its exact status.*

Thus the atlas is a branch-complete structural classification and factorization result. It does not identify different complete parameter, boundary, history or state data as one physical realization.

Proof. The accepted rooted chain and its kind reception give (i); their maps are retained restrictions, not reconstructed from a physical table. Canonical bridge reception and the full structural common-source theorem 6.4, together with Theorem 5.6, give (ii). The source operations are received before their contextual signature is formed. Primitive domain reflection, independent continuation and the context congruence proof give (iii).

Theorem 11.2 and the two local reconstruction theorems give (iv). The role table receives every primitive demand; Theorems 8.4, 8.7 and 8.8 compute the exact compatible live image and its occupied fixed-datum closure, including the certified nonlive complement. Corollary 8.11 supplies the preformed positive law core, while Corollary 8.12 retains the exact stronger response domains. They give (v) without replacing a positive occupant by a status entry.

Theorem 10.1 and the projection kernel test give the exact process and projection structure. The marking-torsor and global label results then give (vi). Theorem 11.4 and Propositions 11.5–11.6 give (vii). Each of these constructions receives its source before using its output. Its restriction recovers the old source record and carries no constructor of a missing authority type. The exact branch base and typed failure records therefore give (viii). No argument selects one value of the independent datum b , proving the final distinction. $\square$

12.2 Conservativity and the physical frontier

Proposition 12.2 (No-backflow under atlas restriction). *Deleting the new mixed observations and later projection coordinates recovers the received source atlases and their original theorem domains. A finer successor can distinguish two old observational classes’ representatives, but it cannot change their old restriction values. An authority-preserving successor can populate a previously unformed stronger projection only by supplying its independently required source witness and declaring that new domain; it cannot do so by renaming an existing response.*

Proof. On actual diagrams restriction deletes the added vertices while retaining the full old subdiagram and its maps. The induced map on observational quotients exists by Corollary 4.5. New contexts can refine the kernel, but their deletion returns the old tests with unchanged values and domains. A physical projection is defined only on its authority packet’s domain. Without a witness of that packet type there is no value of the projection to compose with a name. Supplying a new witness is a source extension with new provenance, not a change to the retained earlier absence or unformed status. $\square$

The relevant layers can now be read without conflating their strengths:

Layer	Established structure	Distinction retained
Premetric constraint	Kernel governance; exact source continuation; the AER constraint-necessity result	No extra gravitational particle or carrier, and no numerical metric dynamics selected.
Structural atlas	Kind-positioned roots, one Higgs, complete mixed quotient, role and source-law closure	Distinct boundary and response data remain; terminal physical occupancy is a finer type.
Represented local law	Simultaneous gauge–matter and charged Yukawa operations, finite co-efficient reconstruction, relative-flavor invariant reconstruction	Nonzero operations are not nonzero on-shell fermion histories or quantum states.
Received classical branch	The separately source-attached joint classical gauge–Higgs–gravity–scalar history	The inherited zero fermion fields are not replaced by nonzero classical fermion solutions.
Physical projections	Exact authority-indexed neutral, electroweak, Higgs, color, process and record maps	Pole, confinement, interacting-state and unrestricted common-apex authority keep their own domains and requirements.

This ordering explains how a structural re-presentation can be complete without claiming that all later physical constructions have the same readiness. The fixed-branch classical gravity–matter capstone requires its full source-attached metric and refoliation chain. The unrestricted physical apex additionally requires compatible confinement, unstable-state, interacting-QFT and dynamical-gravity authority on one source. Those targets cannot be reached by multiplying the available marginal projections.

12.3 Conclusion

The structural unification rests on a common source and on the complete uses of that source. Fermionic differentiation, gauge response and Higgs fixation retain their different mechanisms while sharing the incidences required for interaction. The mixed quotient identifies exactly the distinctions preserved by the declared operational target. Reconstruction makes that statement explicit at the source, local-law and relative-flavor levels; role closure identifies the positive incidences that coexist at a complete structural datum.

The conventional particle chart is thus recovered through a typed projection, not used to define the identities it names. The premetric constraint layer states the conditions for coherent standing and continuation in a physical interior. The represented law, physical state and boundary constructions add their respective data through explicit maps. This ordering yields a complete structural endpoint while leaving each stronger physical claim attached to the source conditions that can establish it.

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